

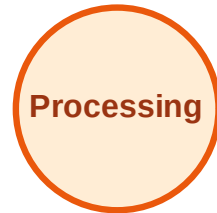
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



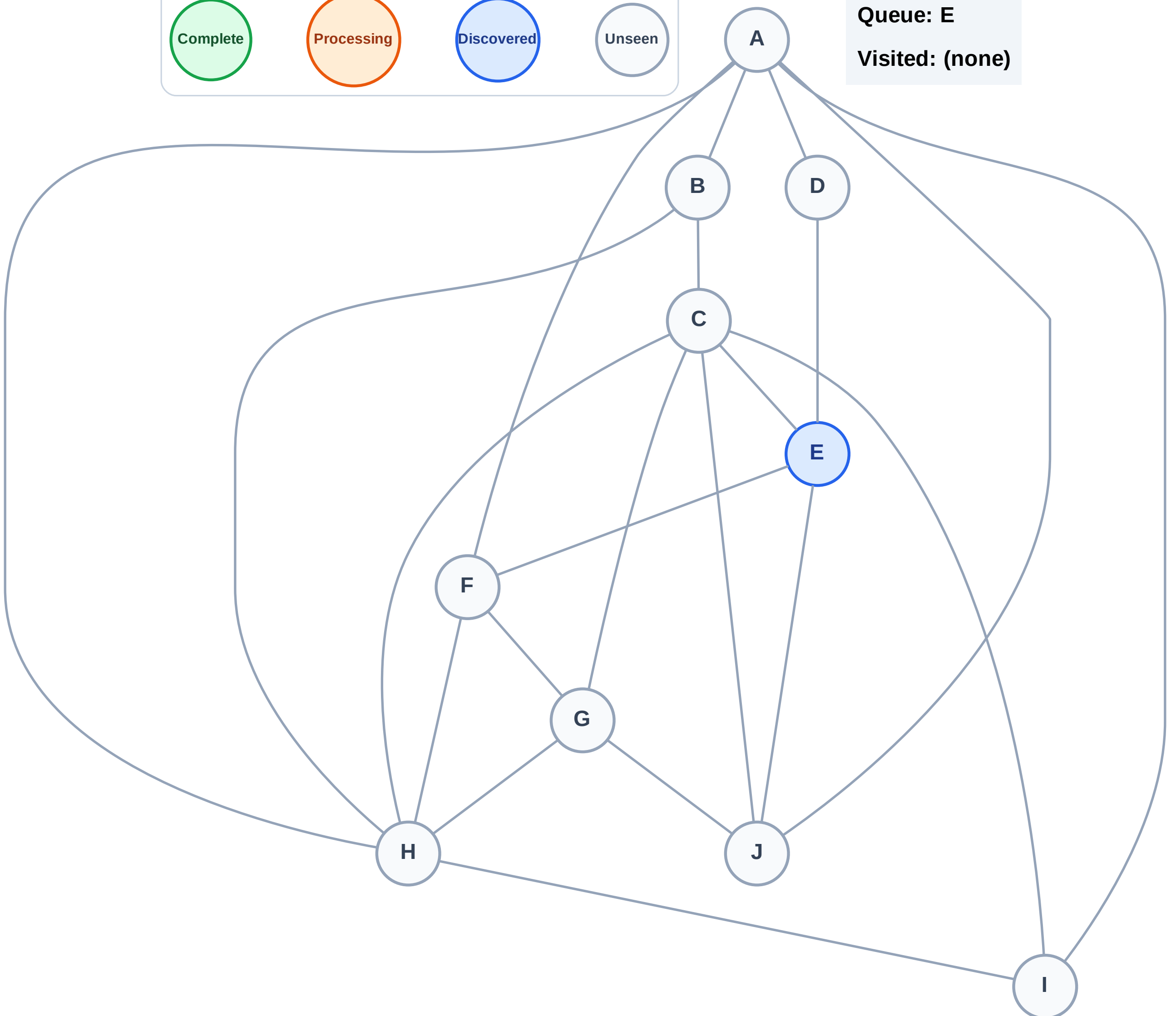
Discovered



Unseen

Queue: E

Visited: (none)



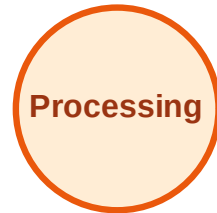
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



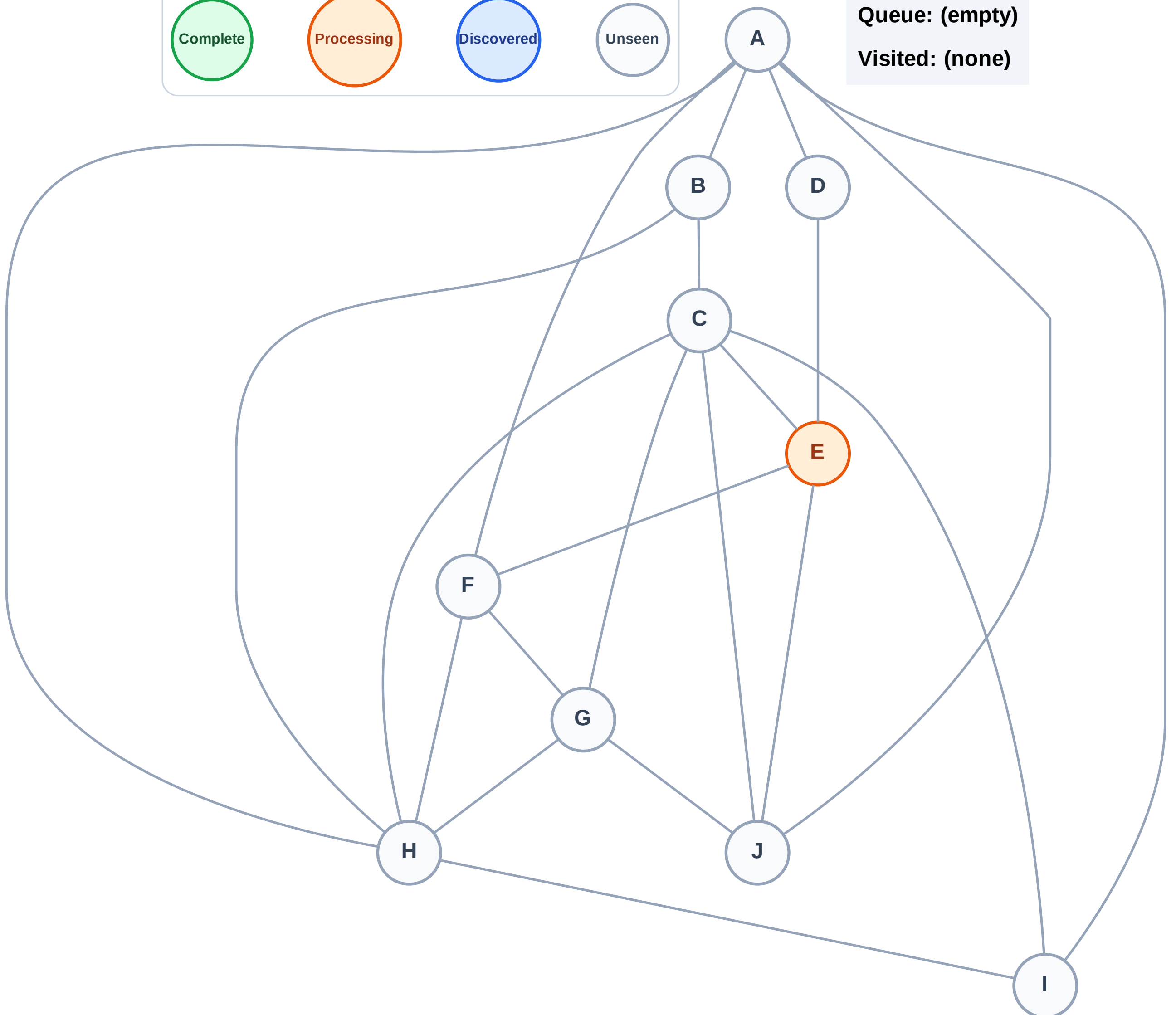
Discovered



Unseen

Queue: (empty)

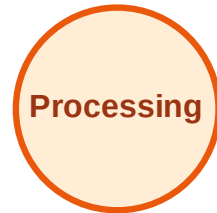
Visited: (none)



BFS Traversal

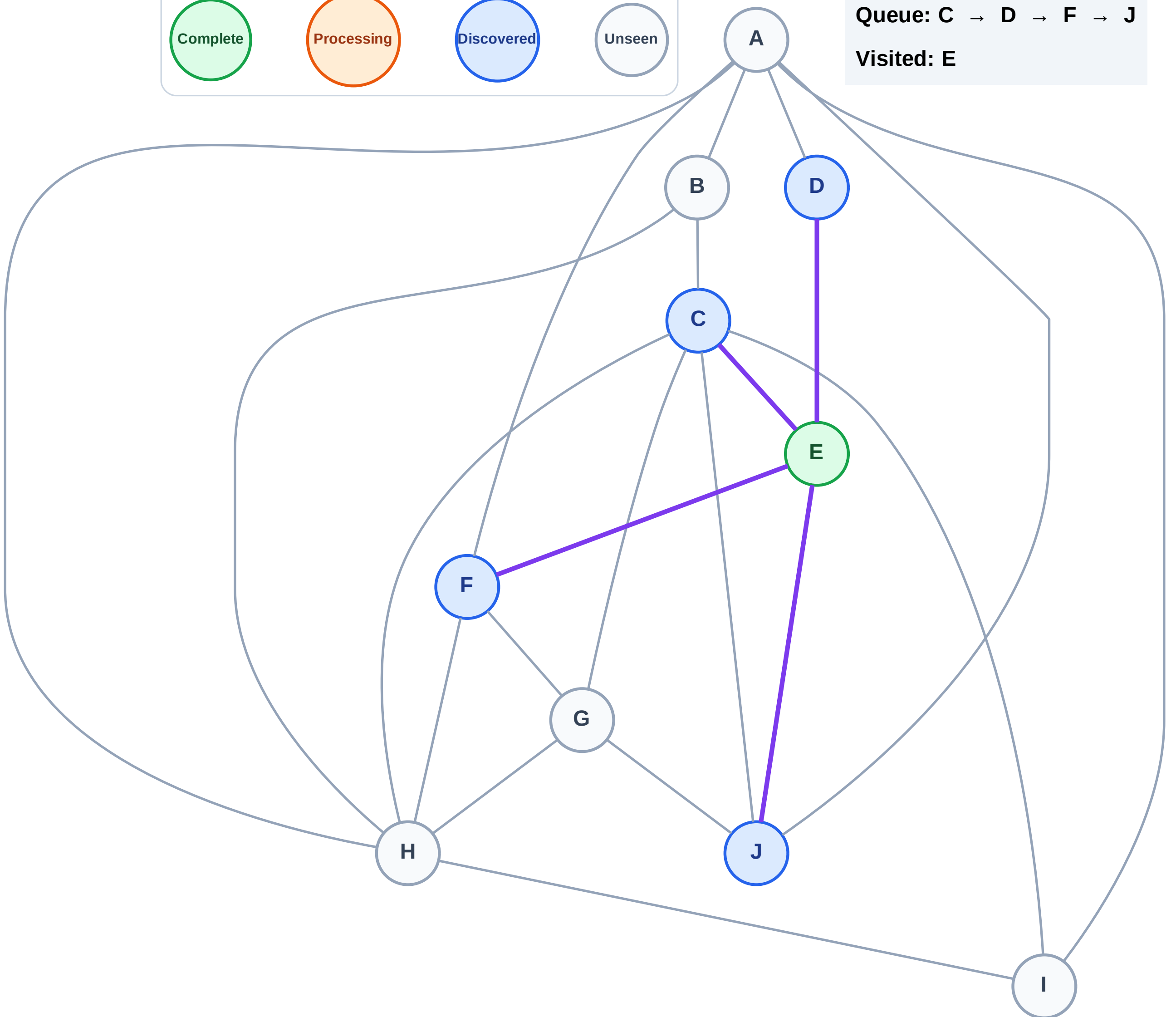
Step 3: Discover C, D, F, J from E

Legend



Queue: C → D → F → J

Visited: E



BFS Traversal

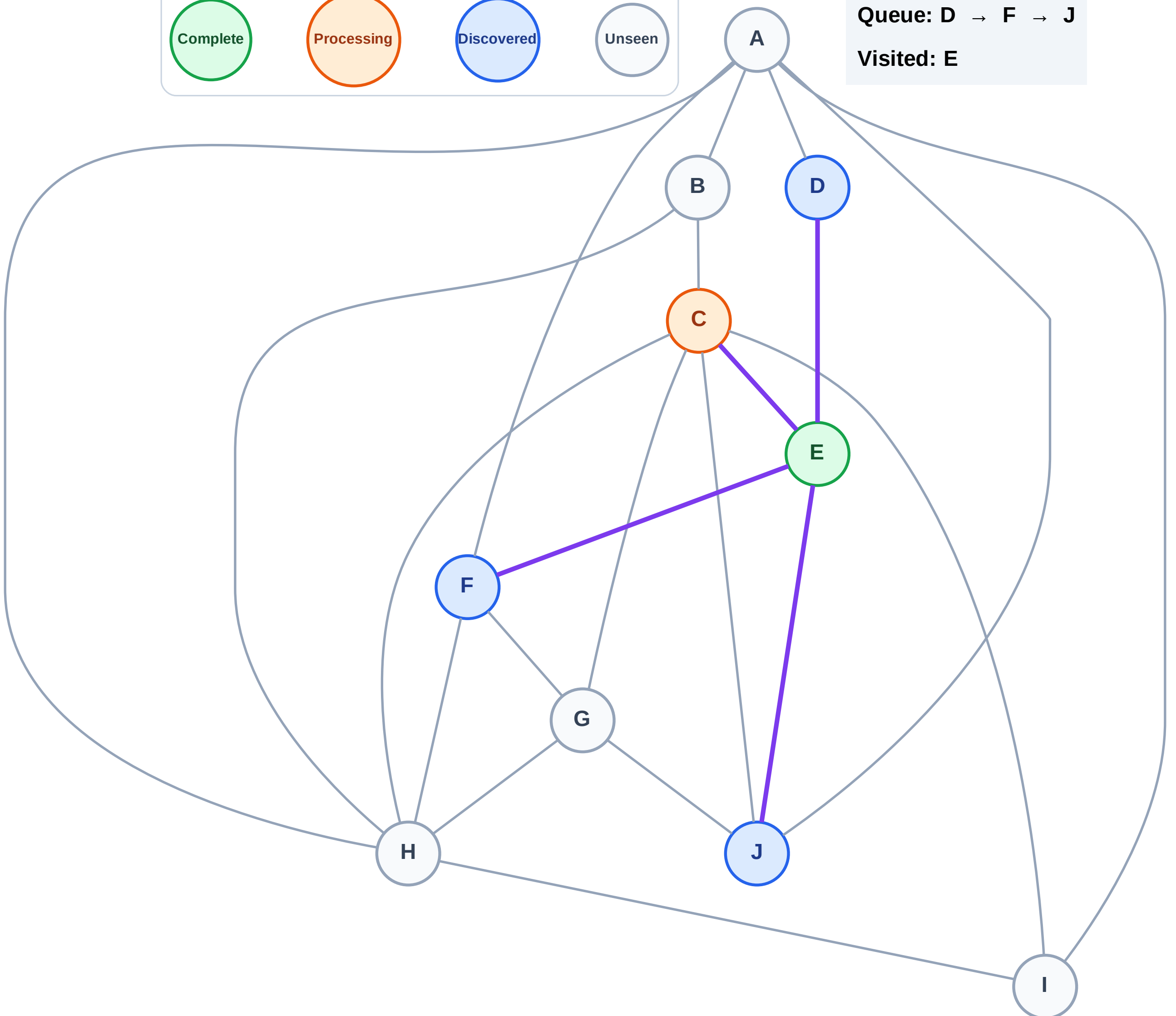
Step 4: Process node C

Legend



Queue: D → F → J

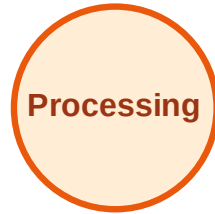
Visited: E



BFS Traversal

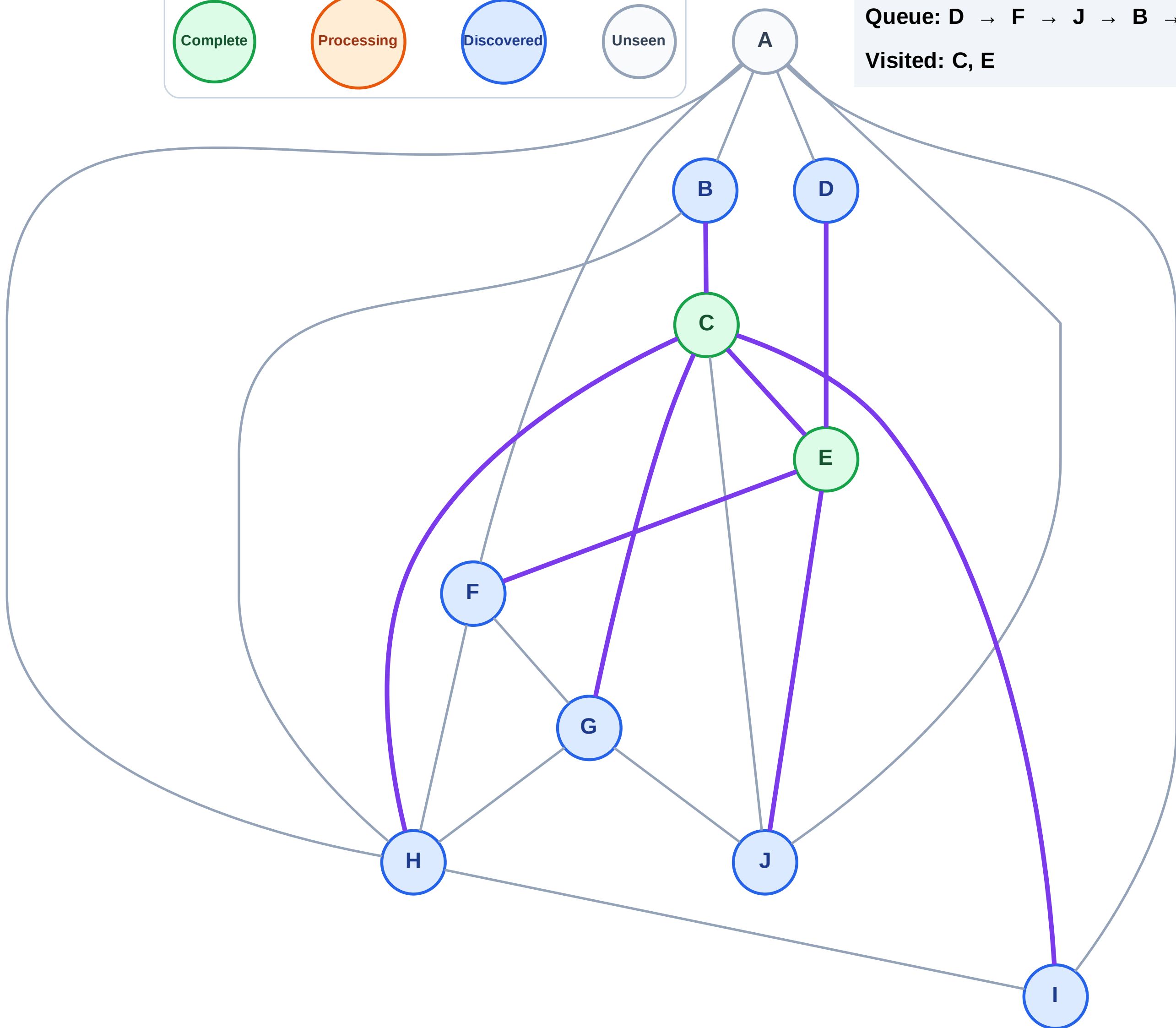
Step 5: Discover B, G, H, I from C

Legend



Queue: D → F → J → B → G → H → I

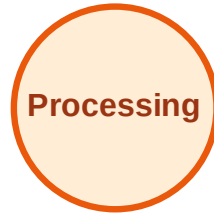
Visited: C, E



BFS Traversal

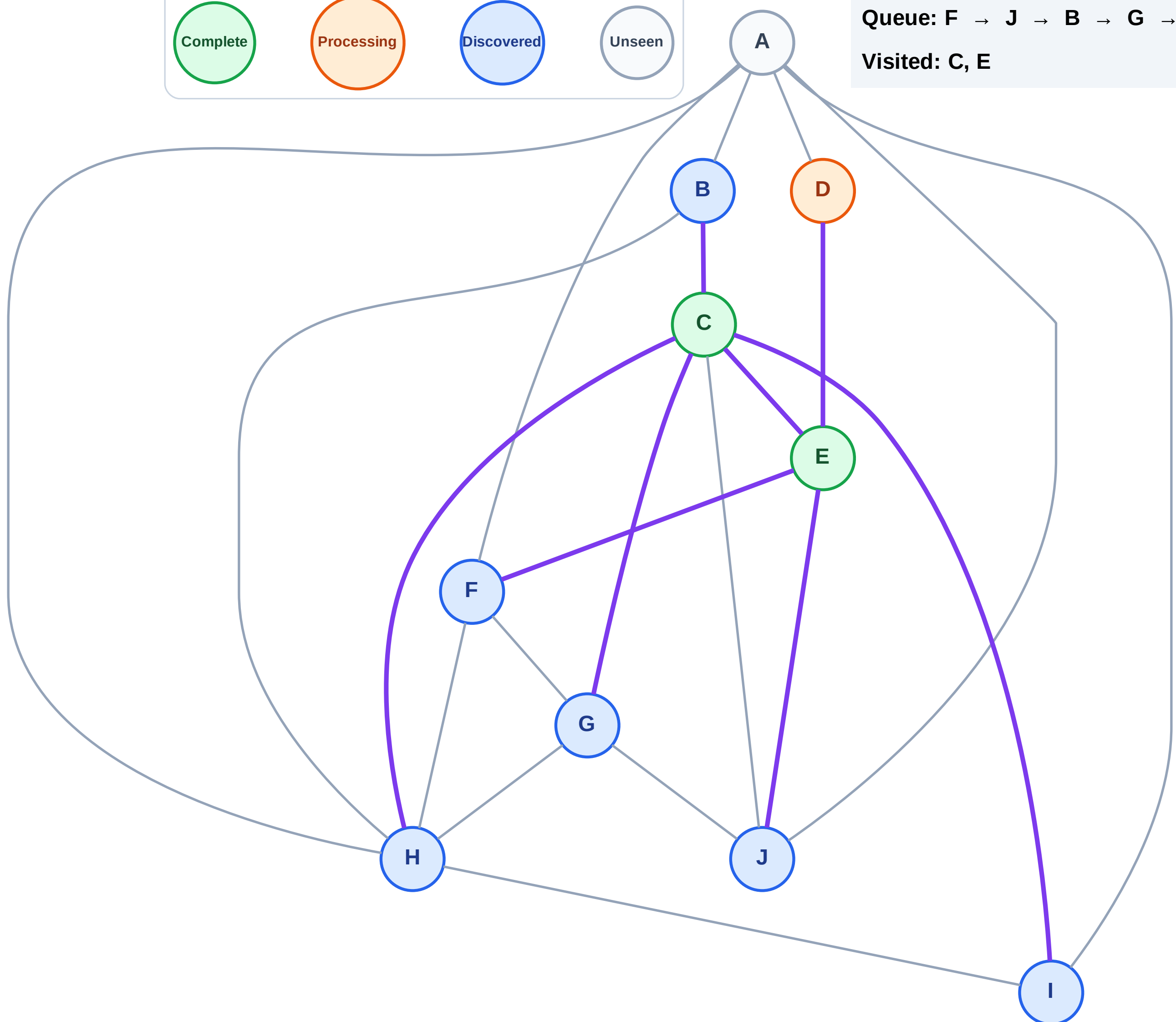
Step 6: Process node D

Legend



Queue: F → J → B → G → H → I

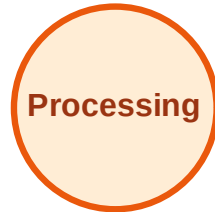
Visited: C, E



BFS Traversal

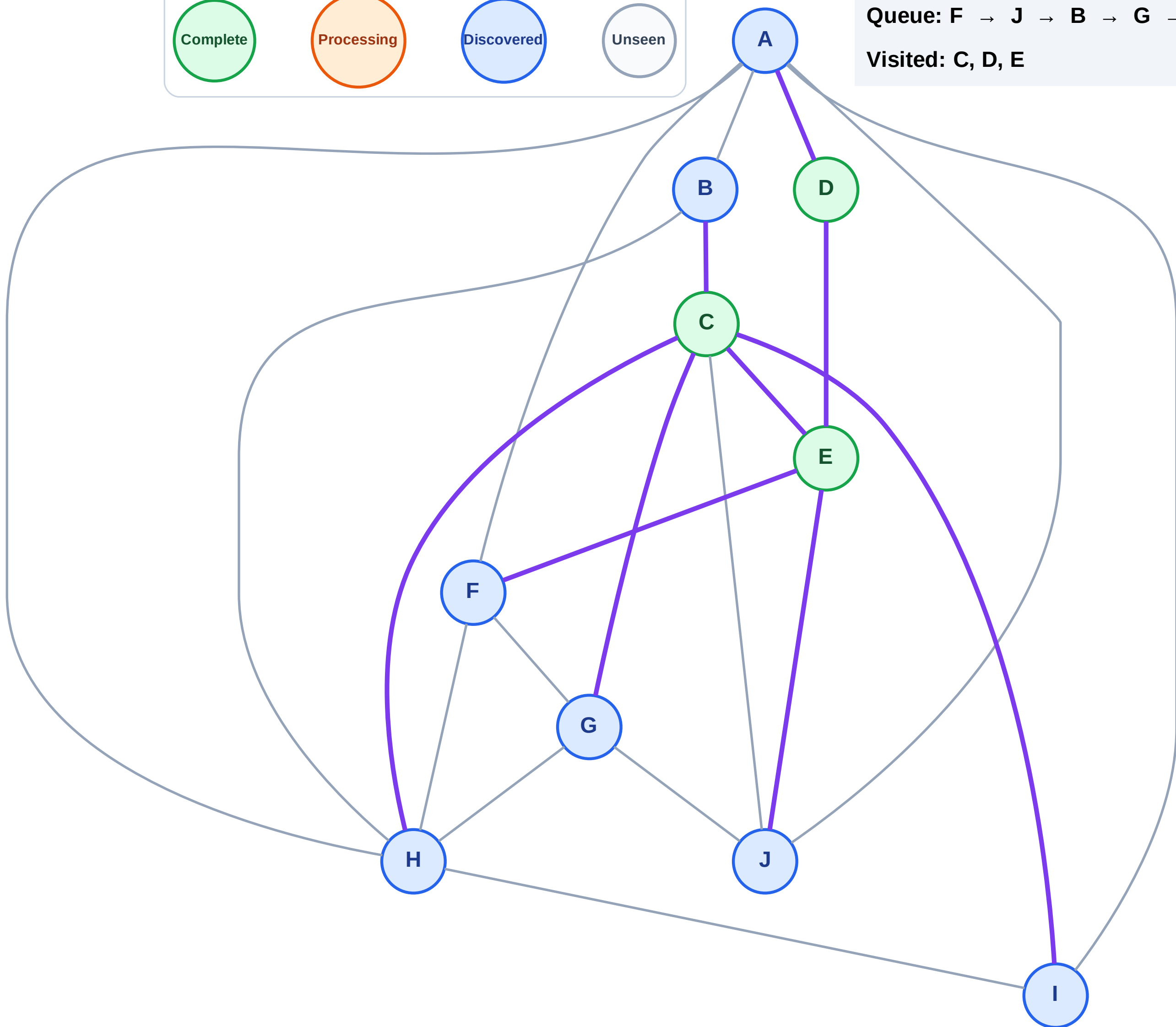
Step 7: Discover A from D

Legend



Queue: F → J → B → G → H → I → A

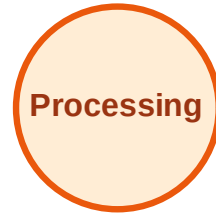
Visited: C, D, E



BFS Traversal

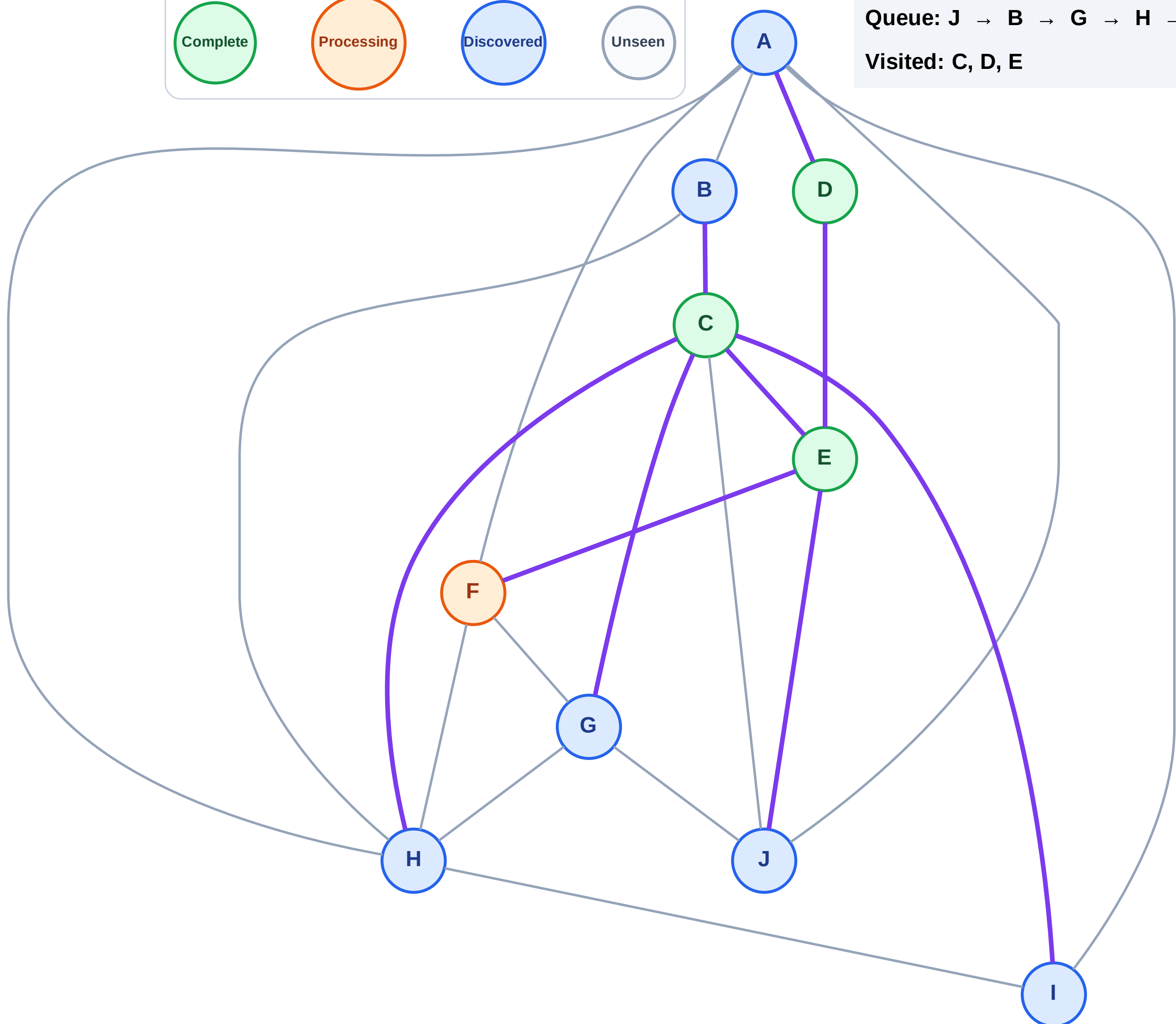
Step 8: Process node F

Legend



Queue: J → B → G → H → I → A

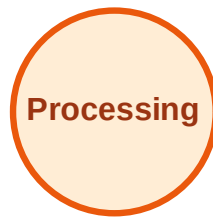
Visited: C, D, E



BFS Traversal

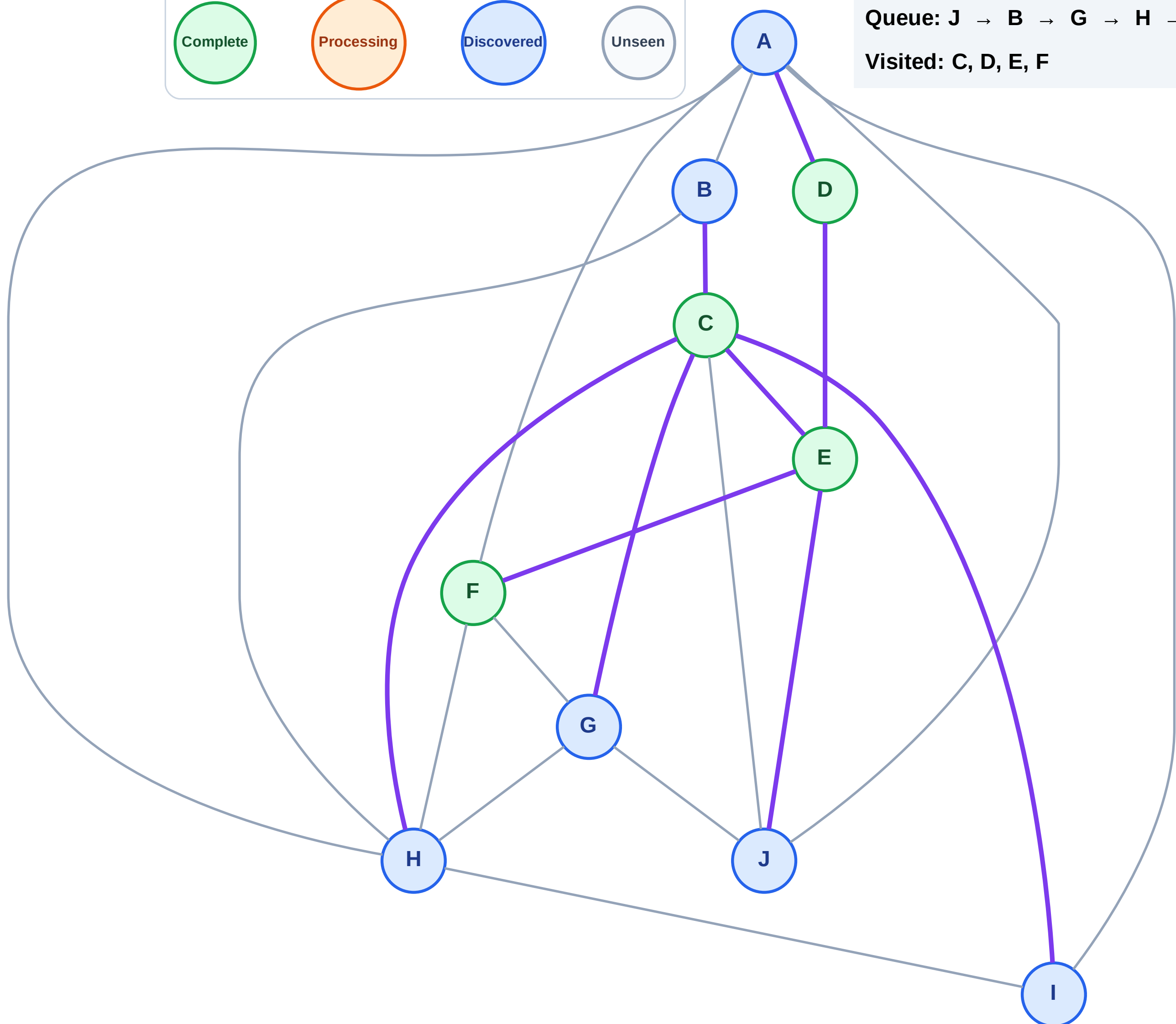
Step 9: No new neighbors from F

Legend



Queue: J → B → G → H → I → A

Visited: C, D, E, F



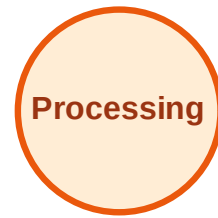
BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



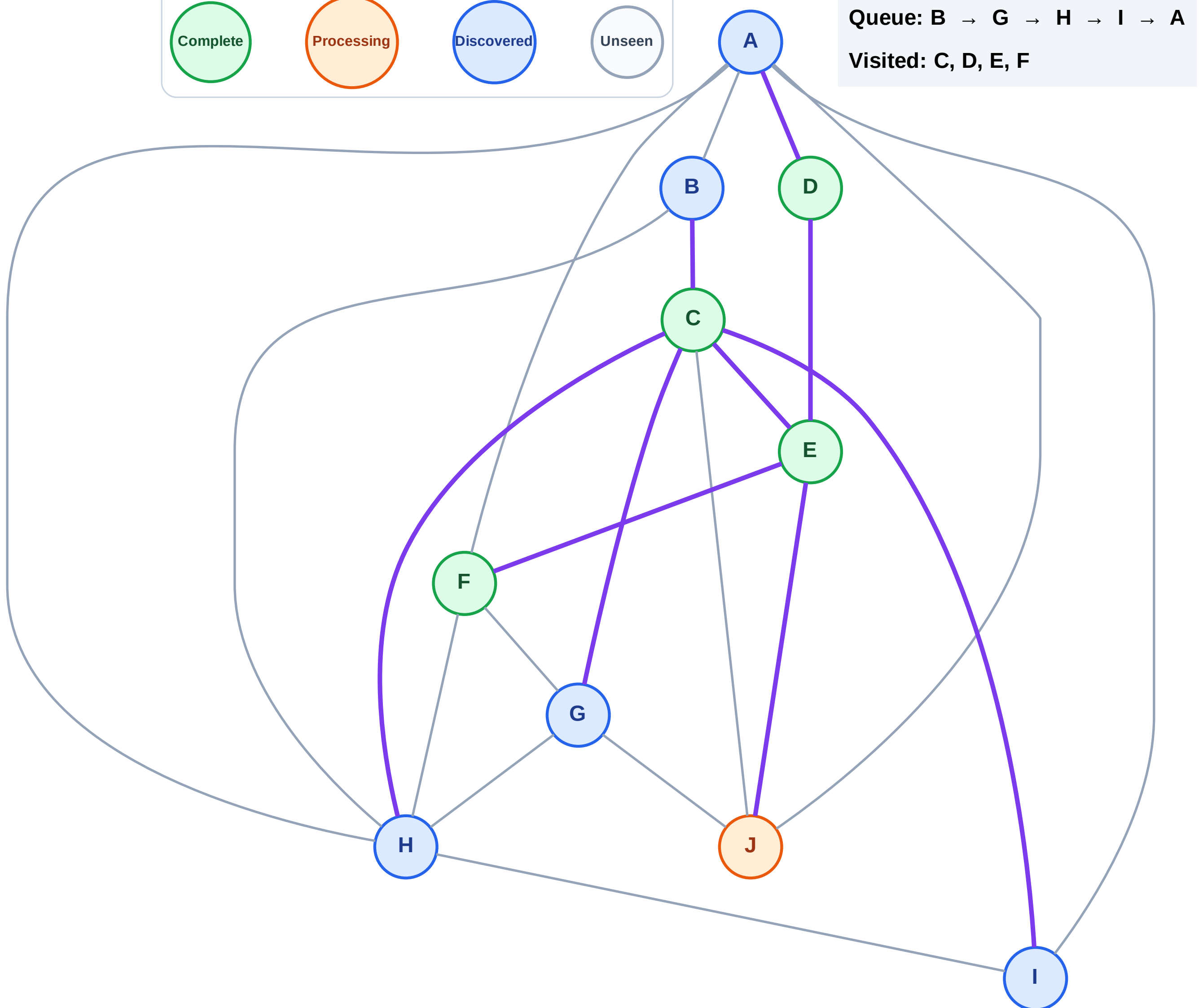
Discovered



Unseen

Queue: B → G → H → I → A

Visited: C, D, E, F



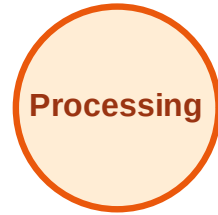
BFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



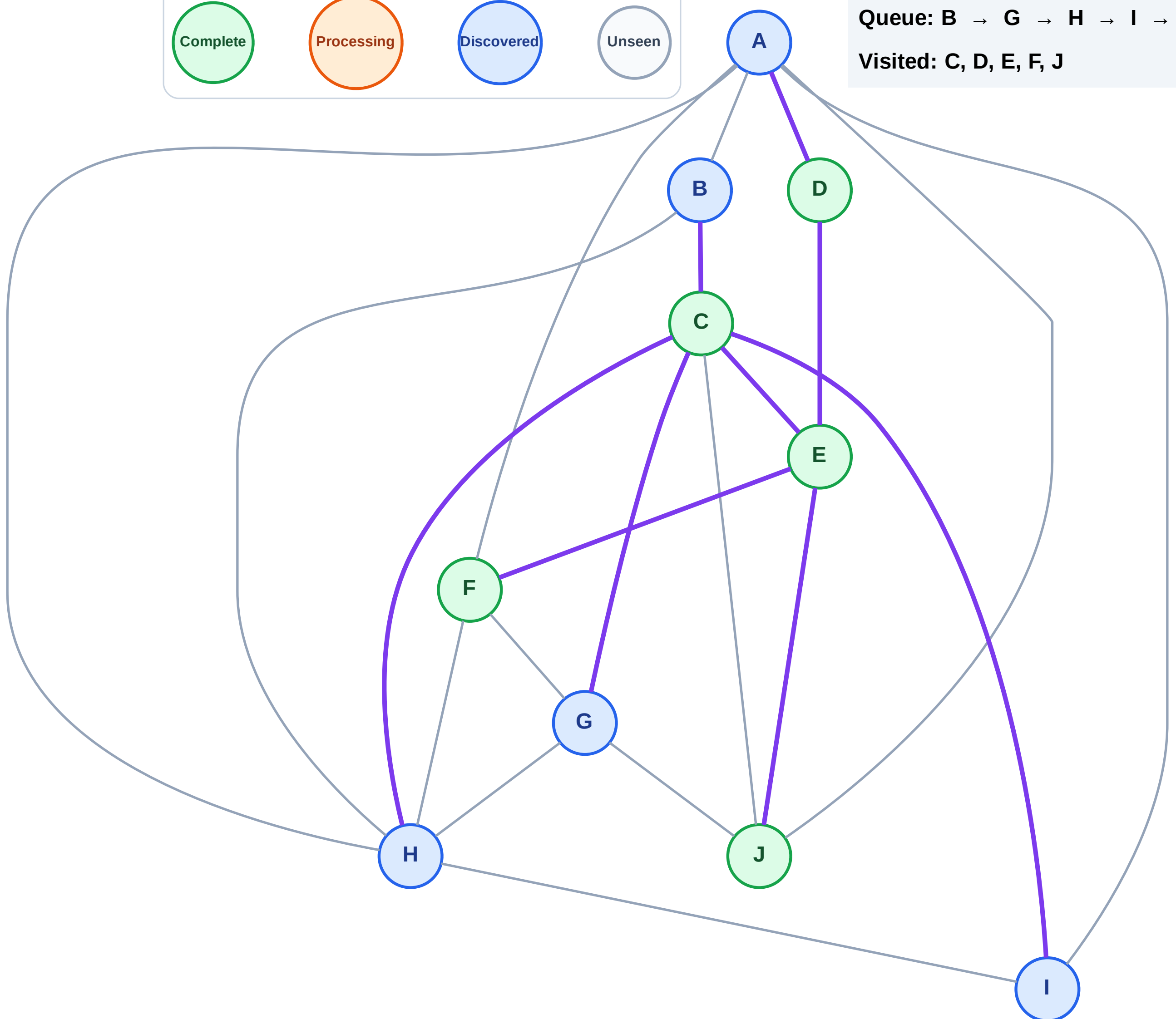
Discovered



Unseen

Queue: B → G → H → I → A

Visited: C, D, E, F, J



BFS Traversal

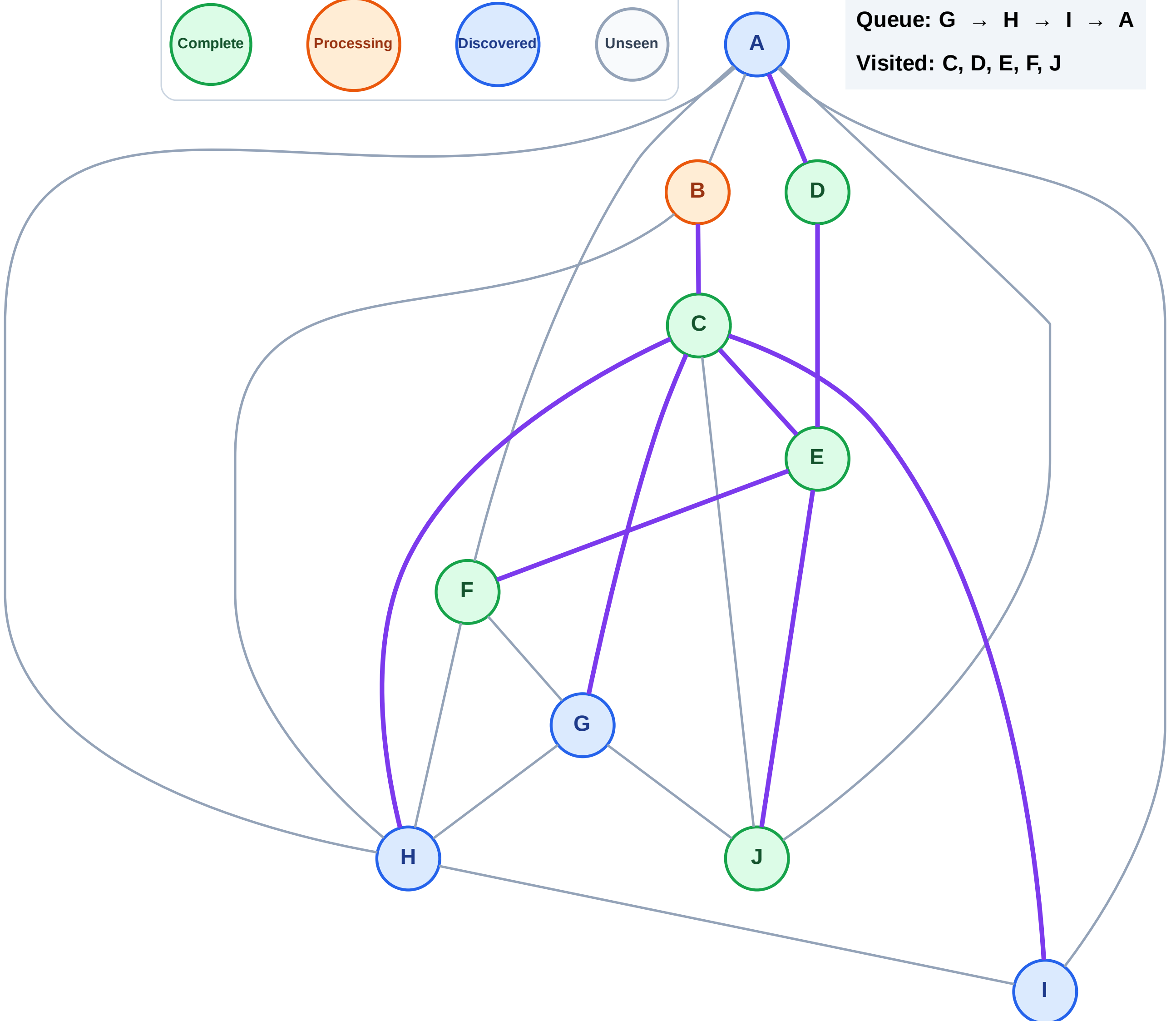
Step 12: Process node B

Legend



Queue: G → H → I → A

Visited: C, D, E, F, J



BFS Traversal

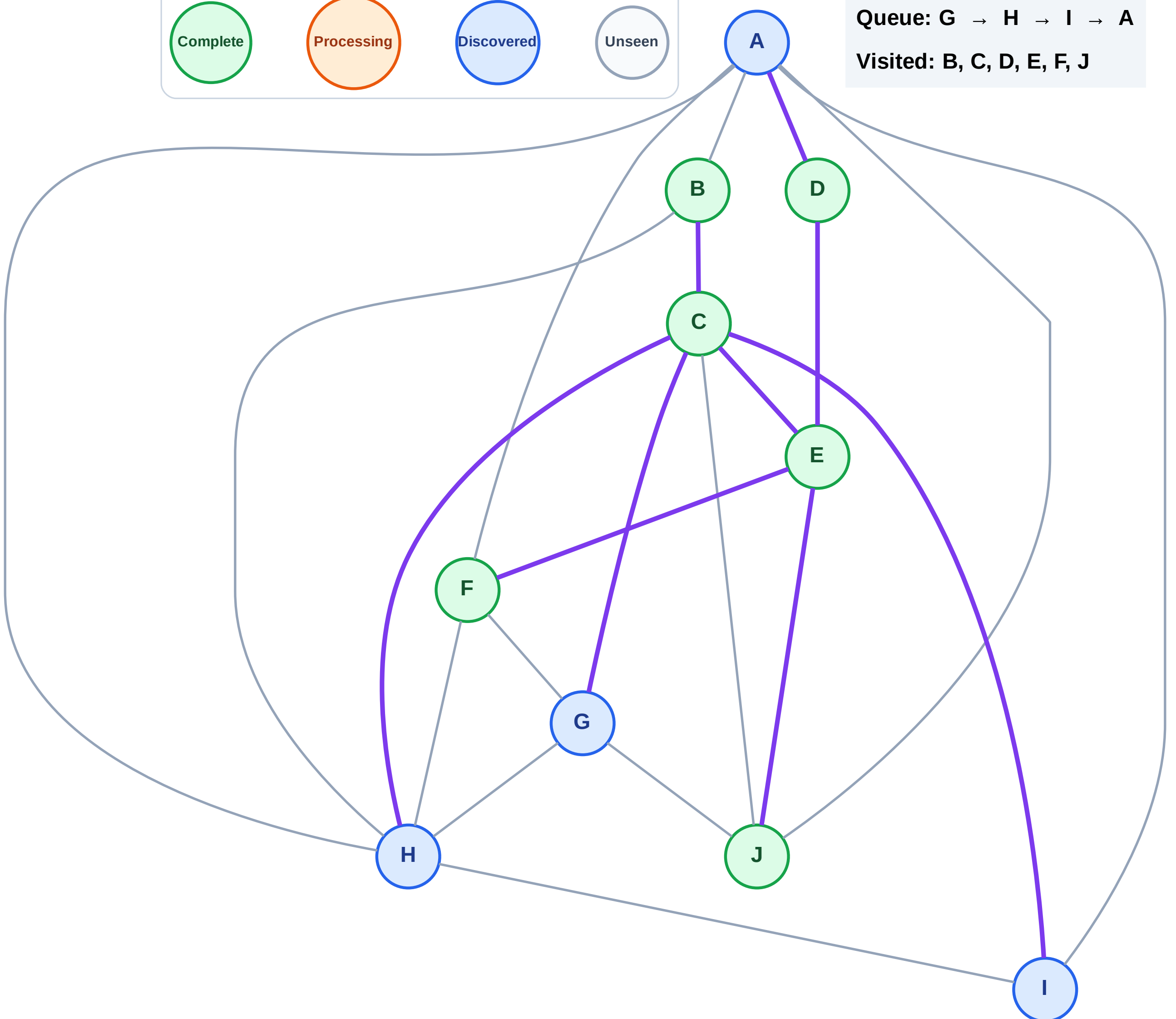
Step 13: No new neighbors from B

Legend



Queue: G → H → I → A

Visited: B, C, D, E, F, J



BFS Traversal

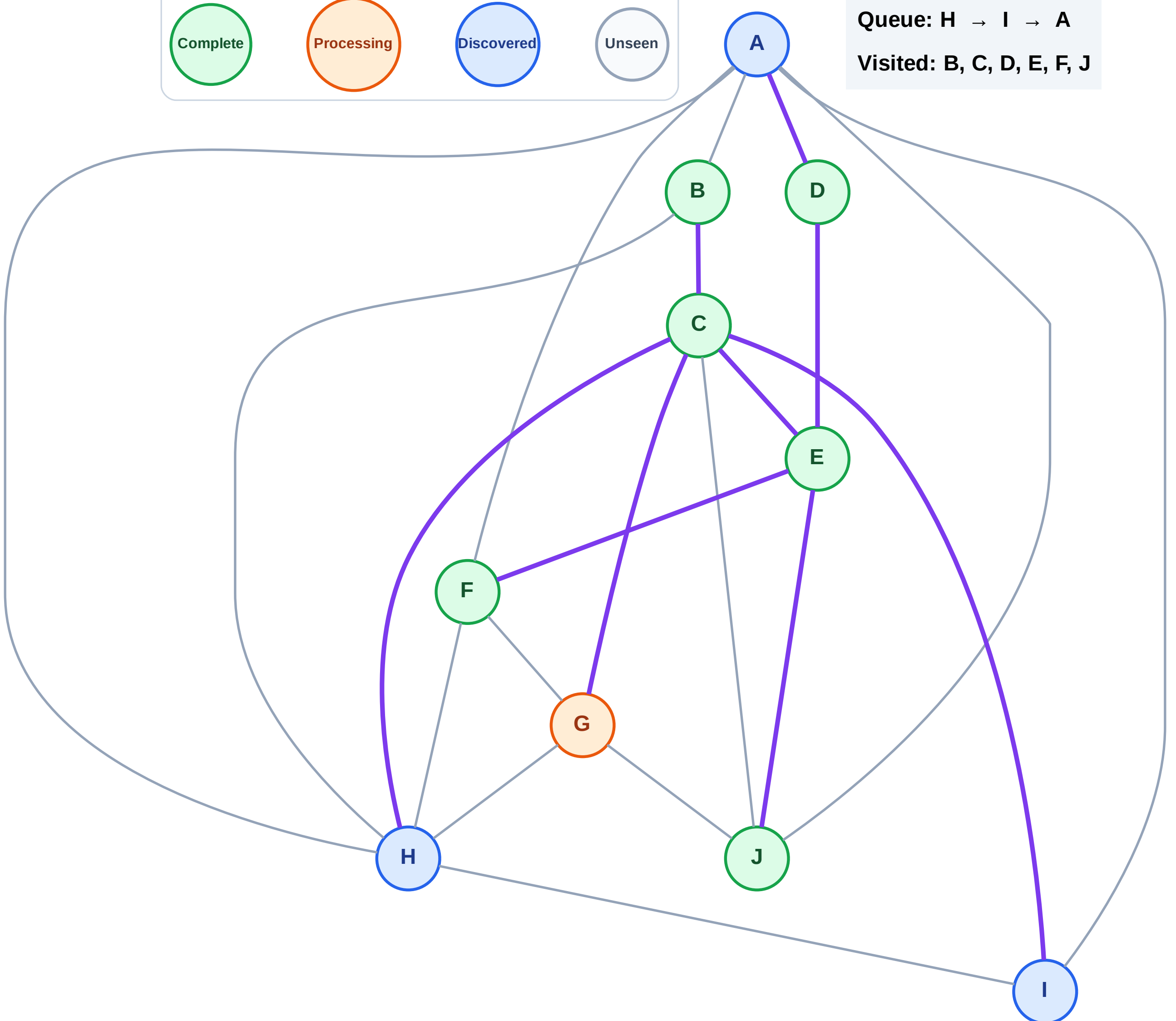
Step 14: Process node G

Legend



Queue: H → I → A

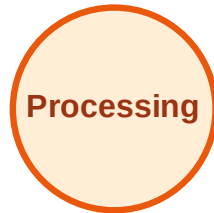
Visited: B, C, D, E, F, J



BFS Traversal

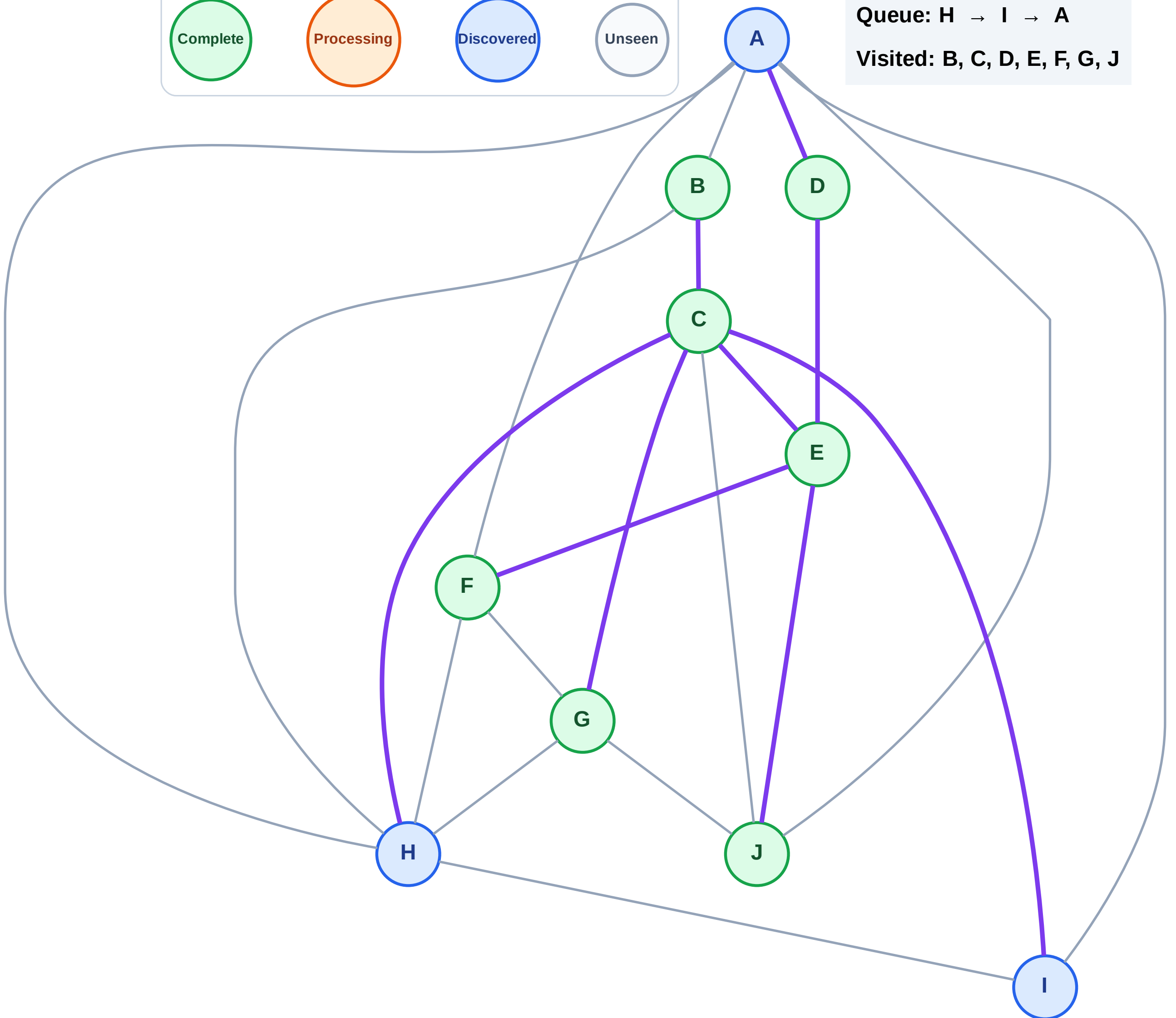
Step 15: No new neighbors from G

Legend



Queue: H → I → A

Visited: B, C, D, E, F, G, J



BFS Traversal

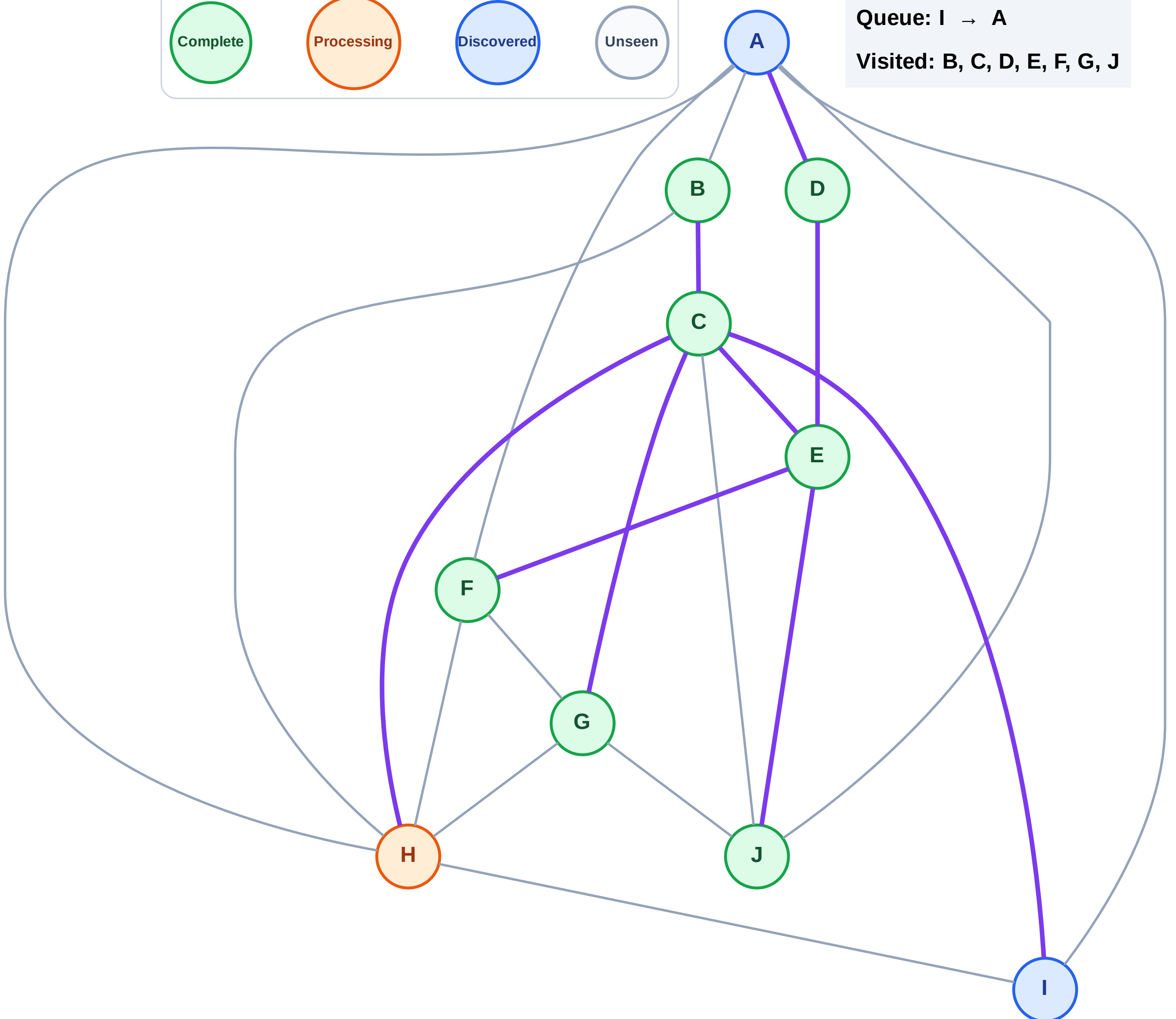
Step 16: Process node H

Legend



Queue: I → A

Visited: B, C, D, E, F, G, J



BFS Traversal

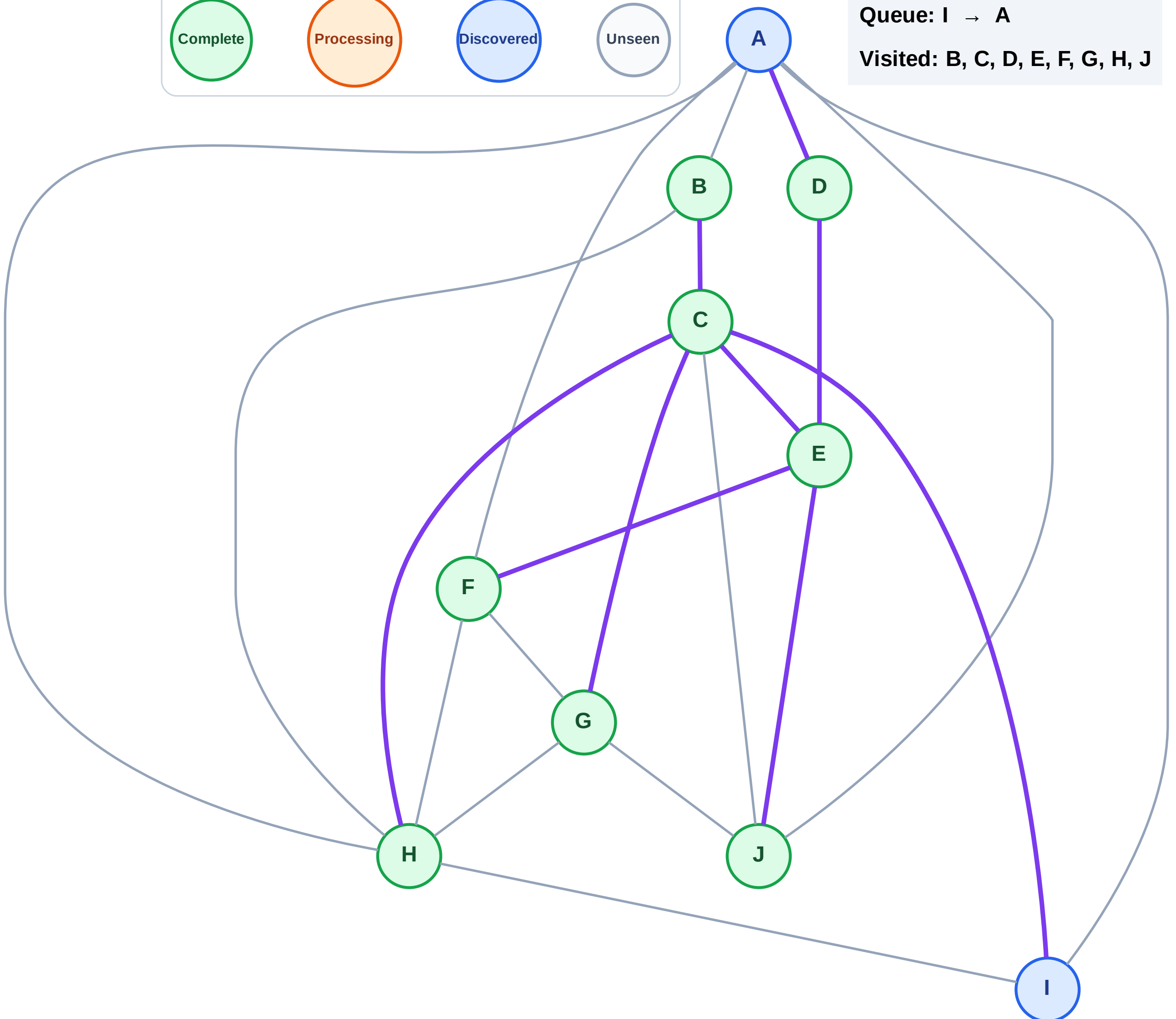
Step 17: No new neighbors from H

Legend



Queue: I → A

Visited: B, C, D, E, F, G, H, J



BFS Traversal

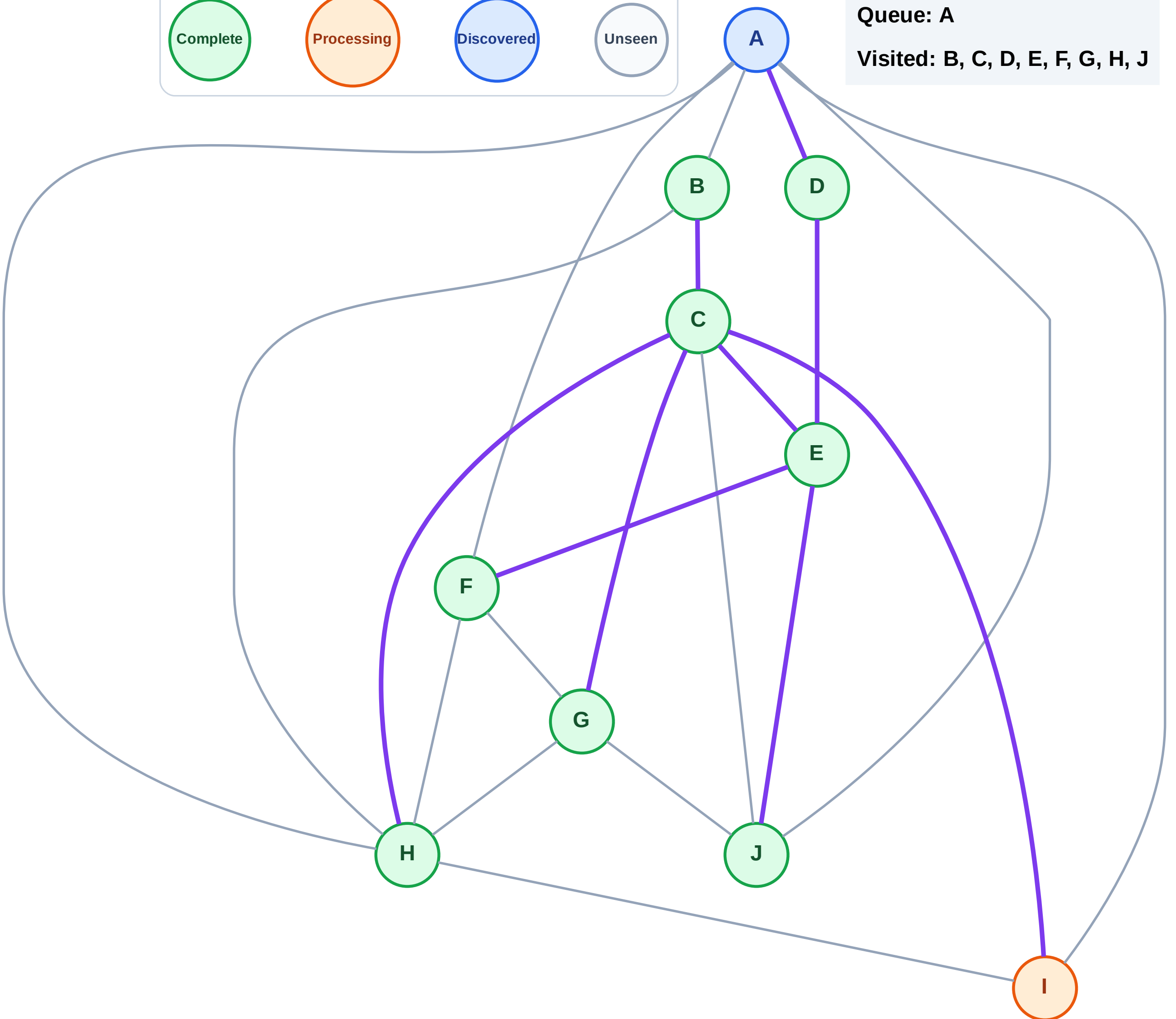
Step 18: Process node I

Legend



Queue: A

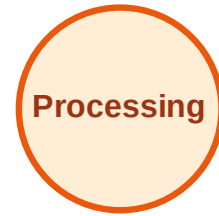
Visited: B, C, D, E, F, G, H, J



BFS Traversal

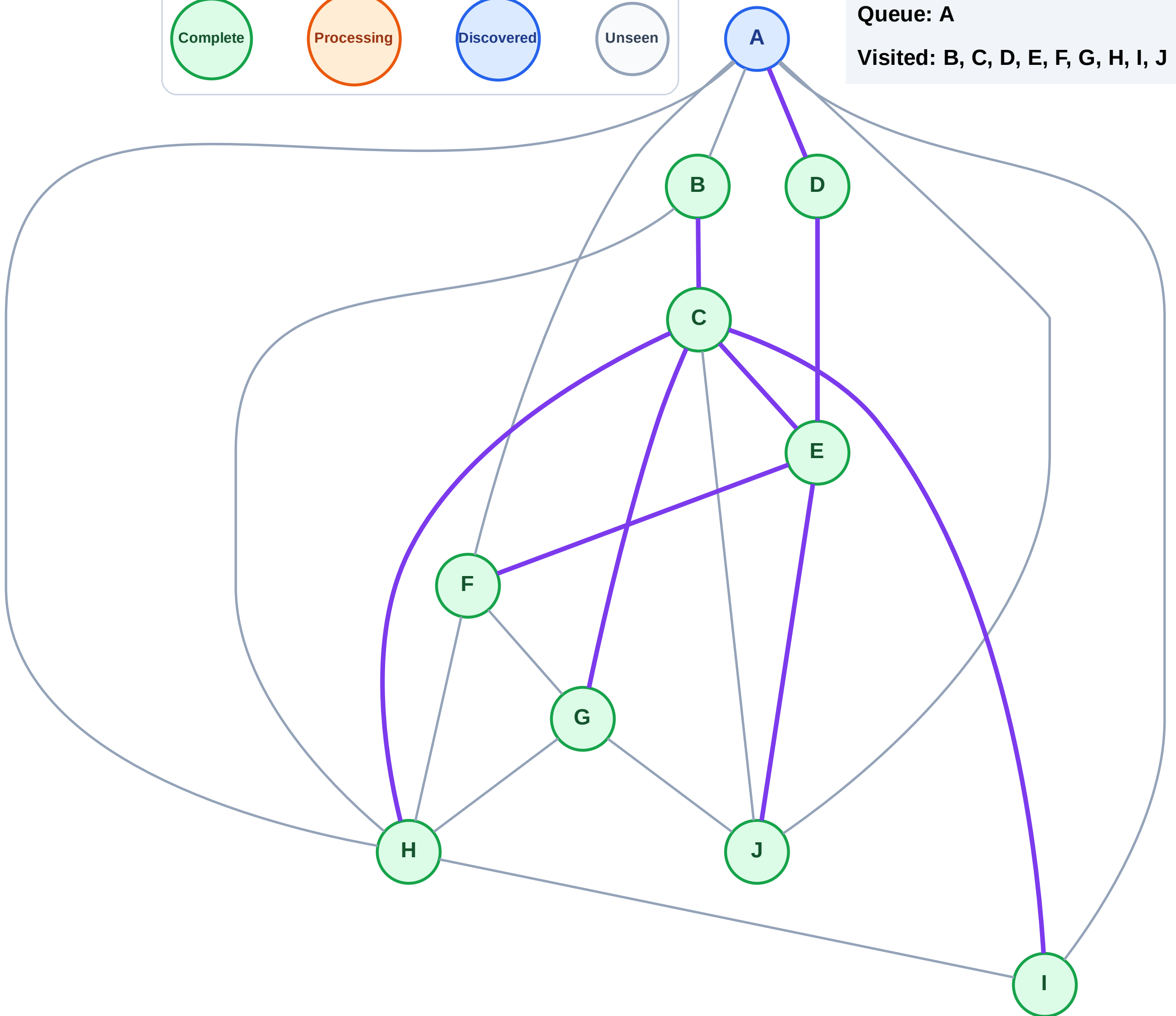
Step 19: No new neighbors from I

Legend



Queue: A

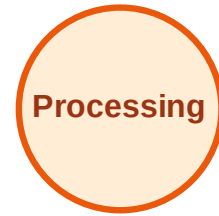
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

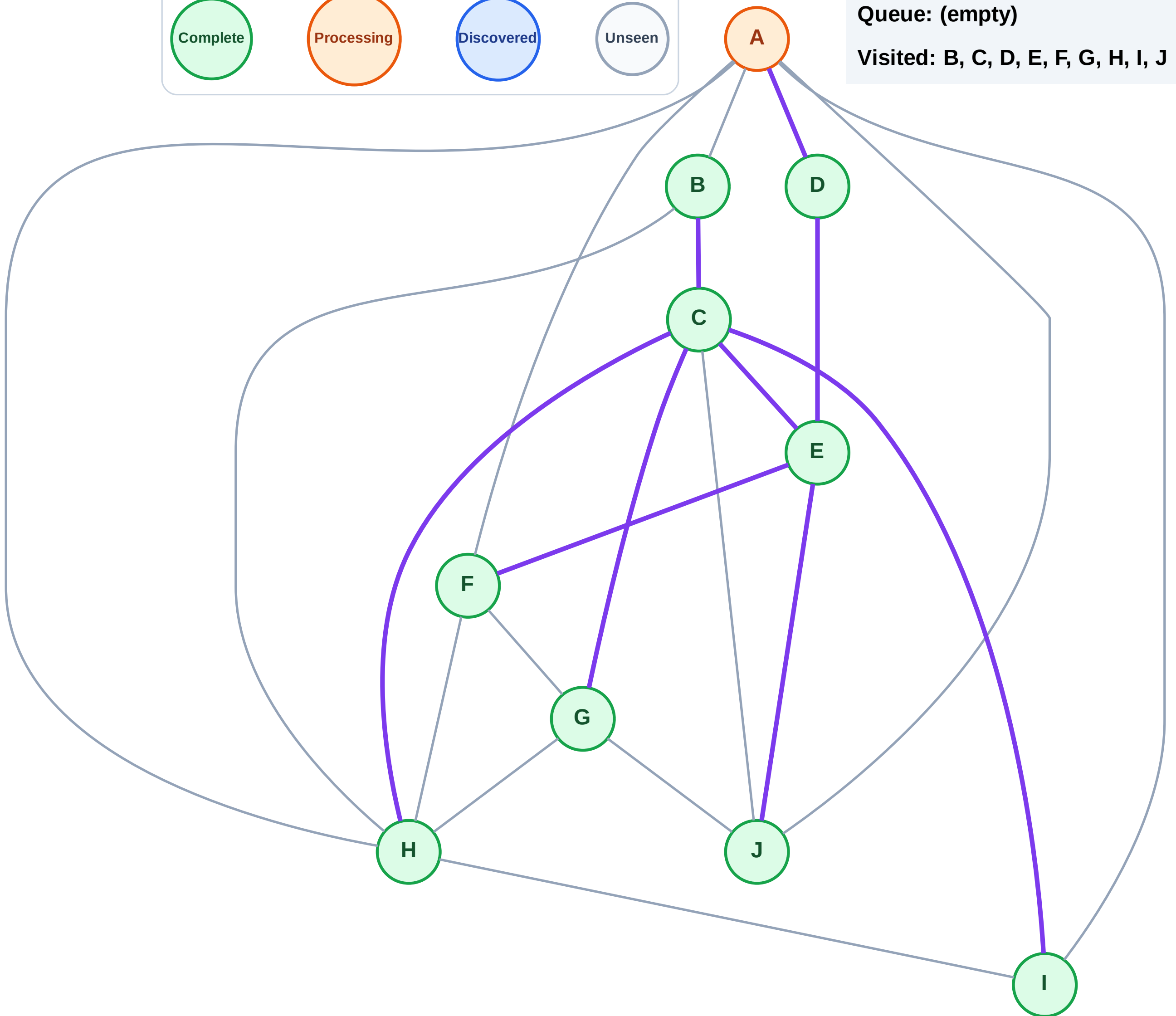
Step 20: Process node A

Legend



Queue: (empty)

Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

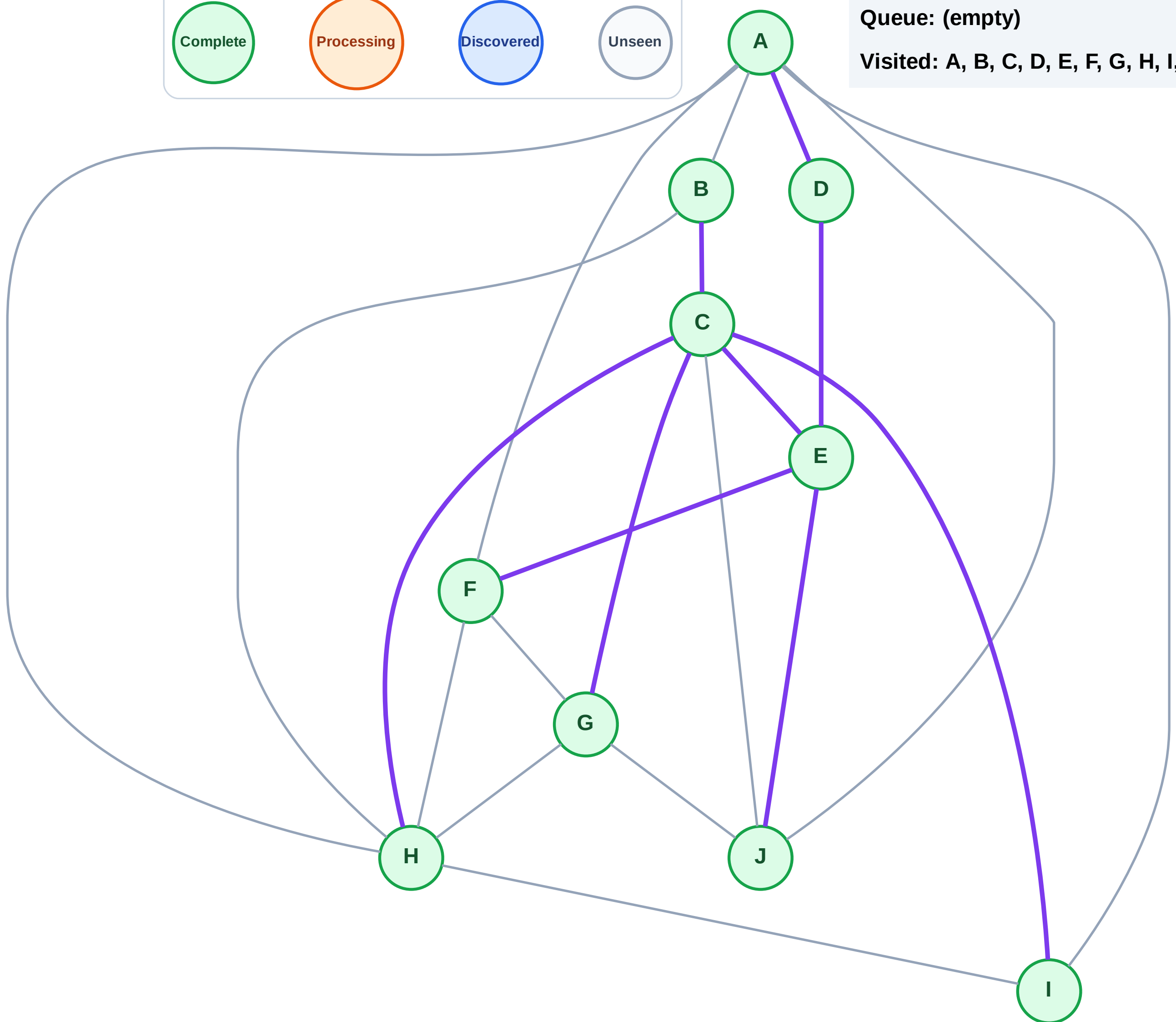
Step 21: No new neighbors from A

Legend



Queue: (empty)

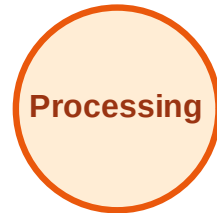
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

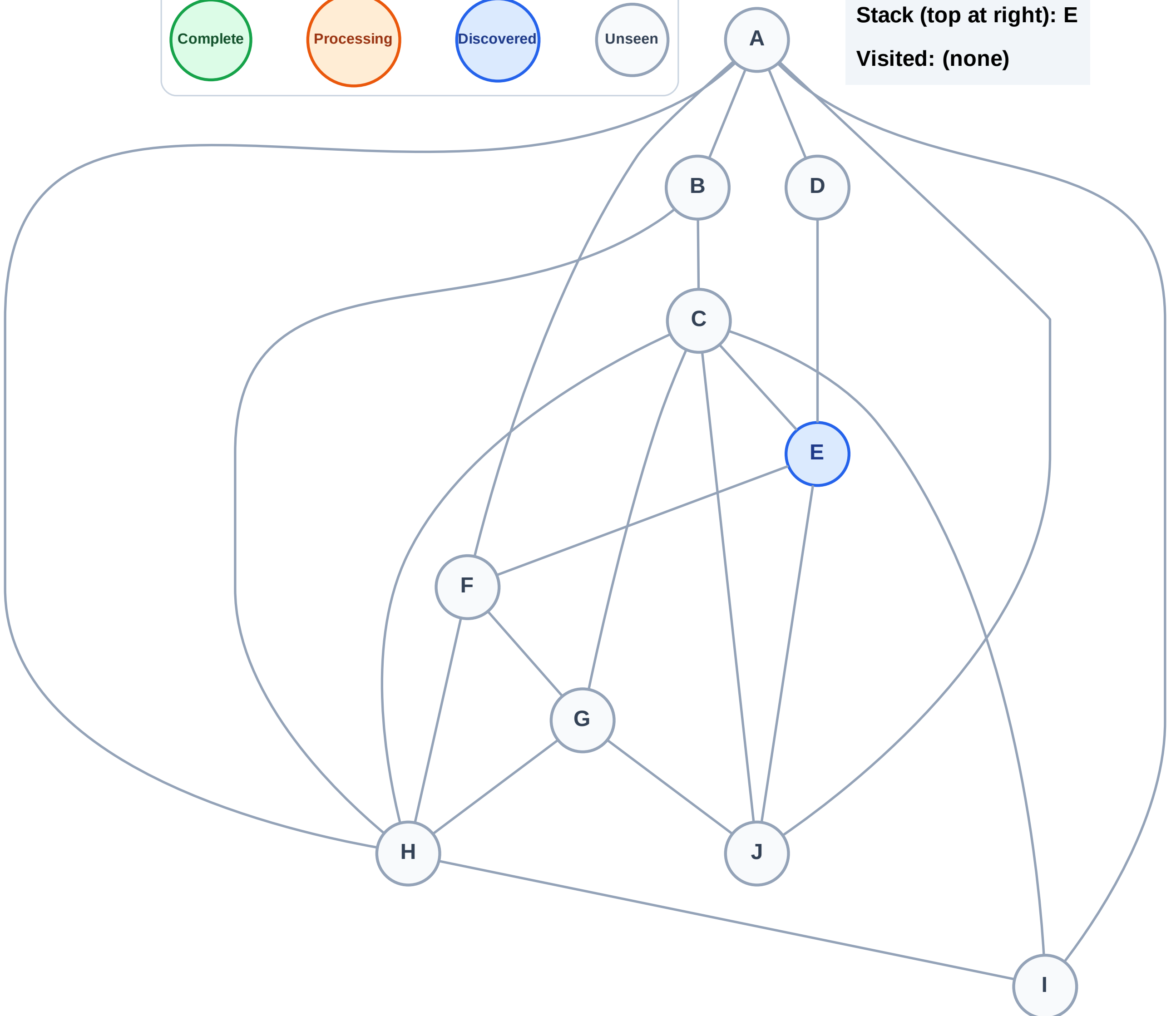
Step 1: Start at E

Legend



Stack (top at right): E

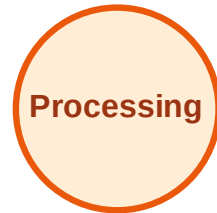
Visited: (none)



DFS Traversal

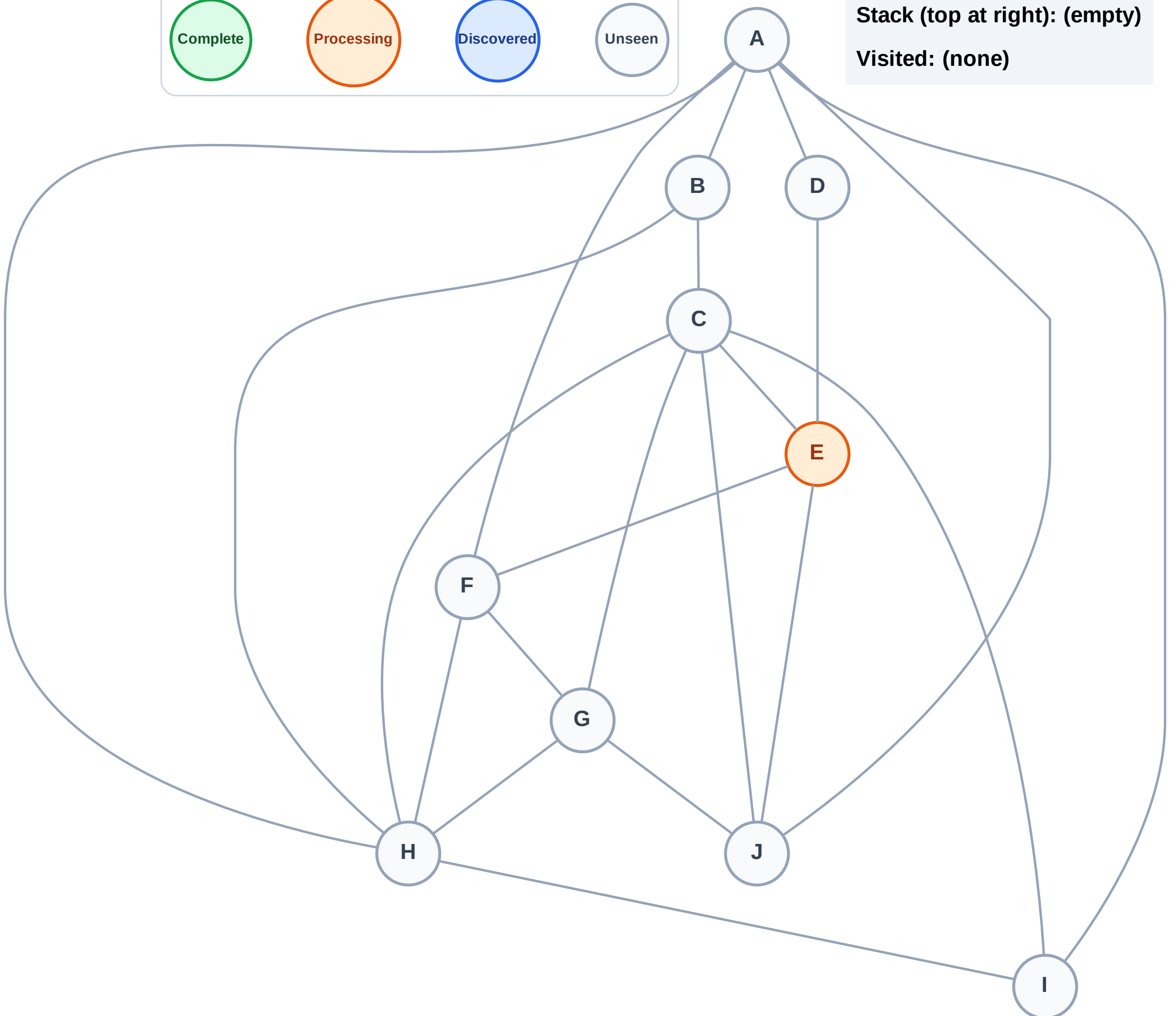
Step 2: Process node E

Legend



Stack (top at right): (empty)

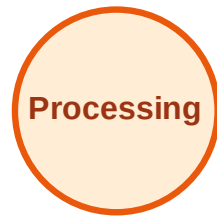
Visited: (none)



DFS Traversal

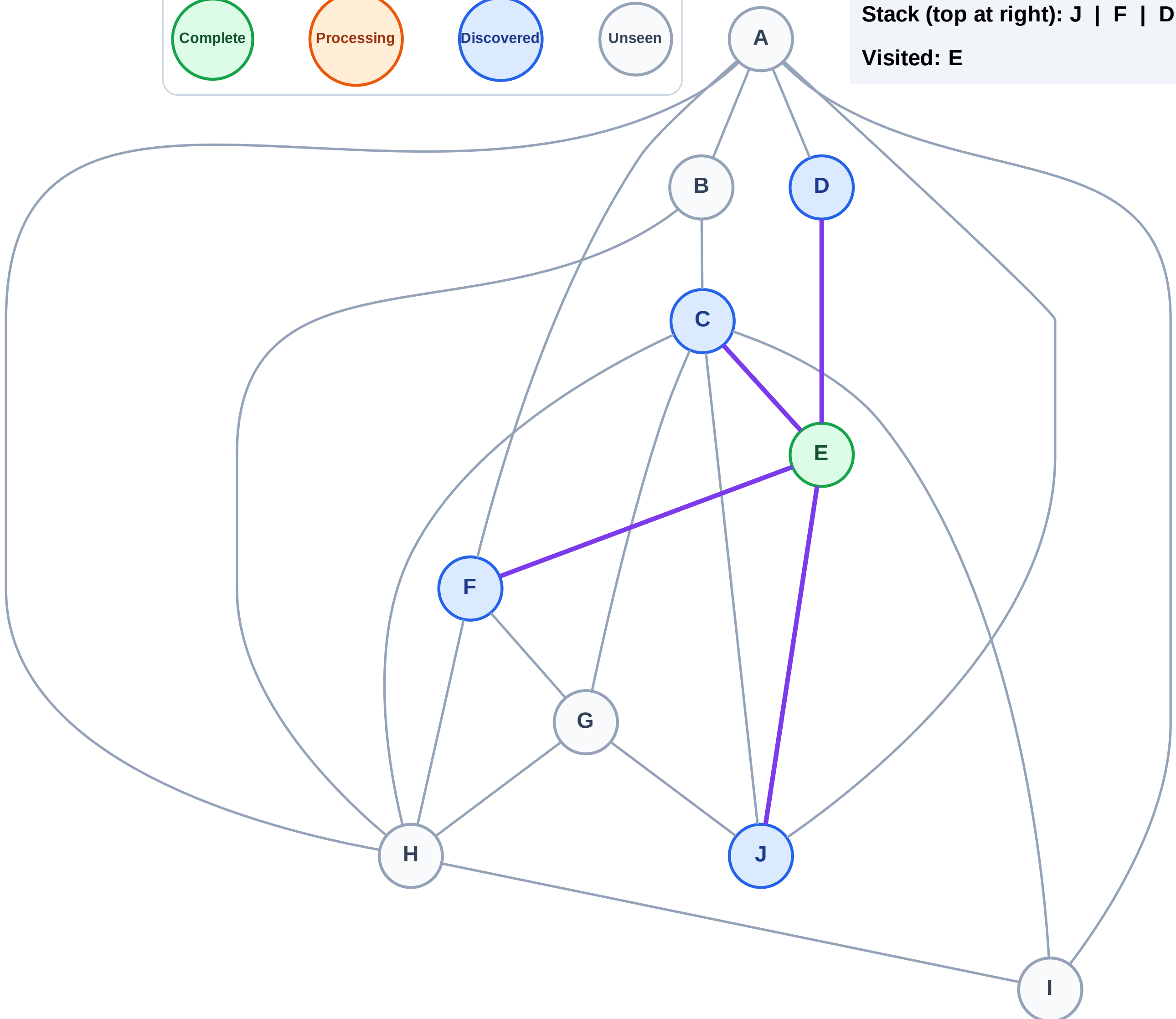
Step 3: Discover J, F, D, C from E

Legend



Stack (top at right): J | F | D | C

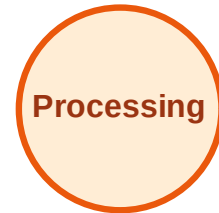
Visited: E



DFS Traversal

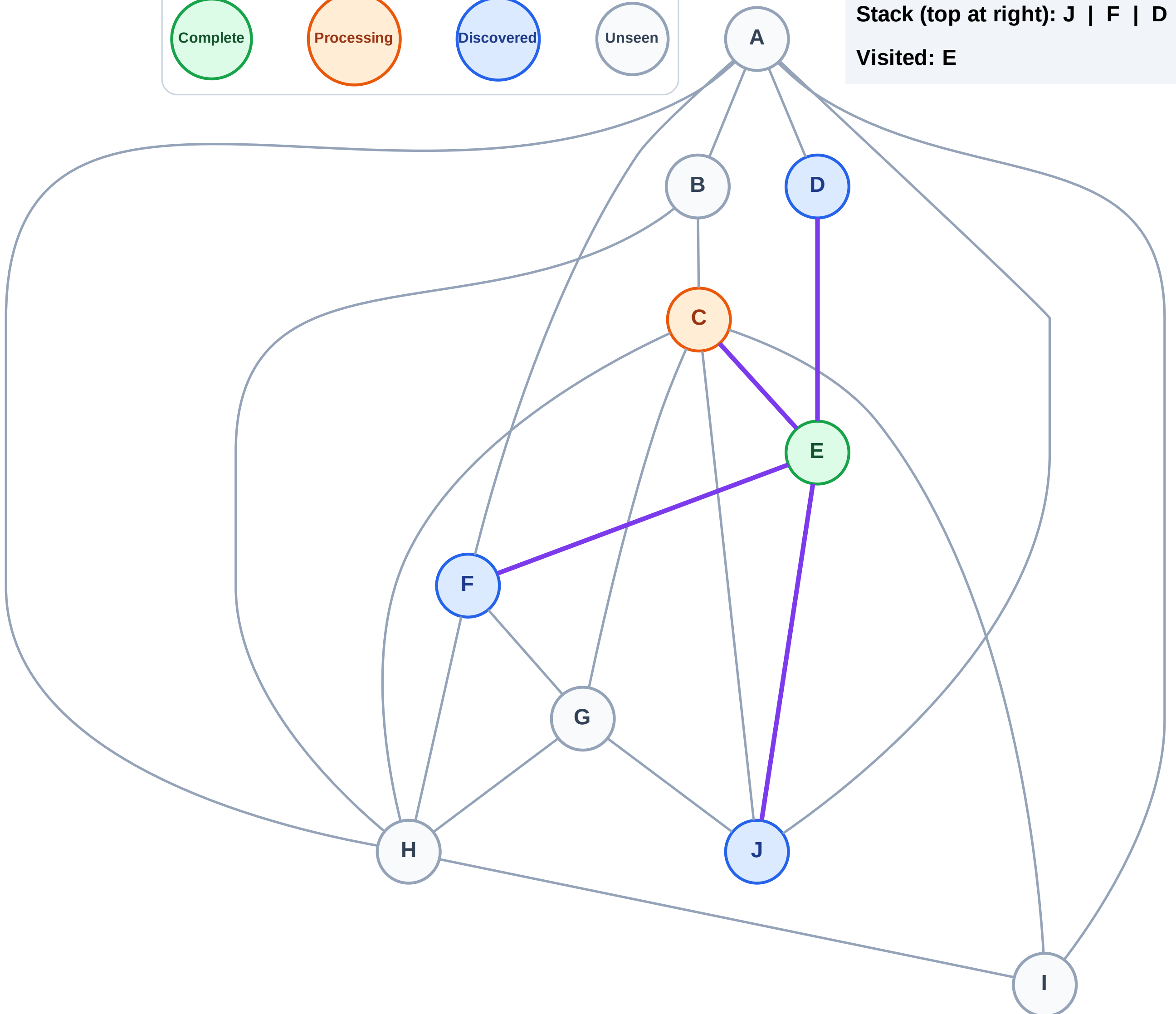
Step 4: Process node C

Legend



Stack (top at right): J | F | D

Visited: E



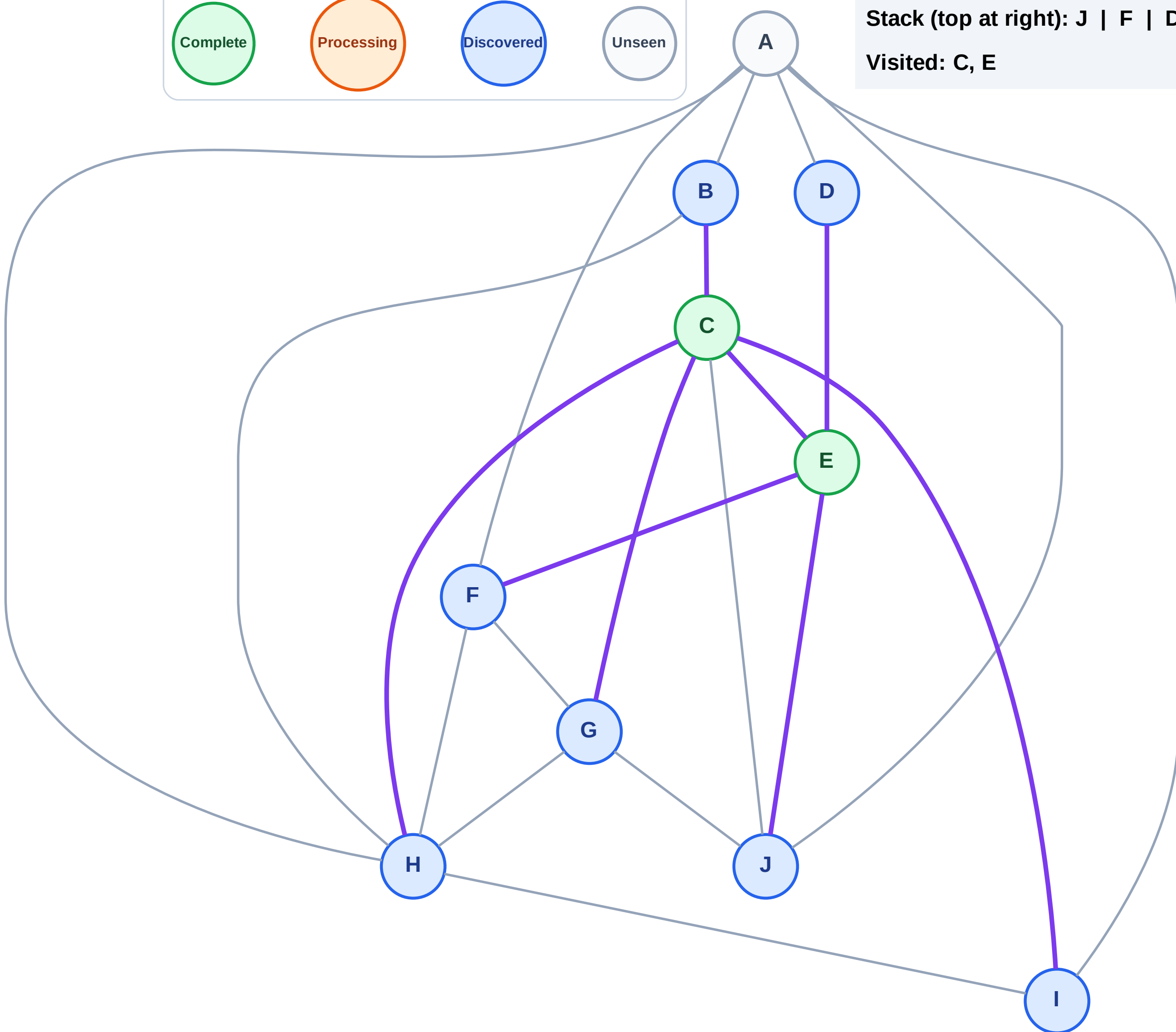
DFS Traversal

Step 5: Discover I, H, G, B from C

Legend



Stack (top at right): J | F | D | I | H | G | B
Visited: C, E



Step 6: Process node B

Step 6: Process node B



Complete

Complete



Processing

Processing



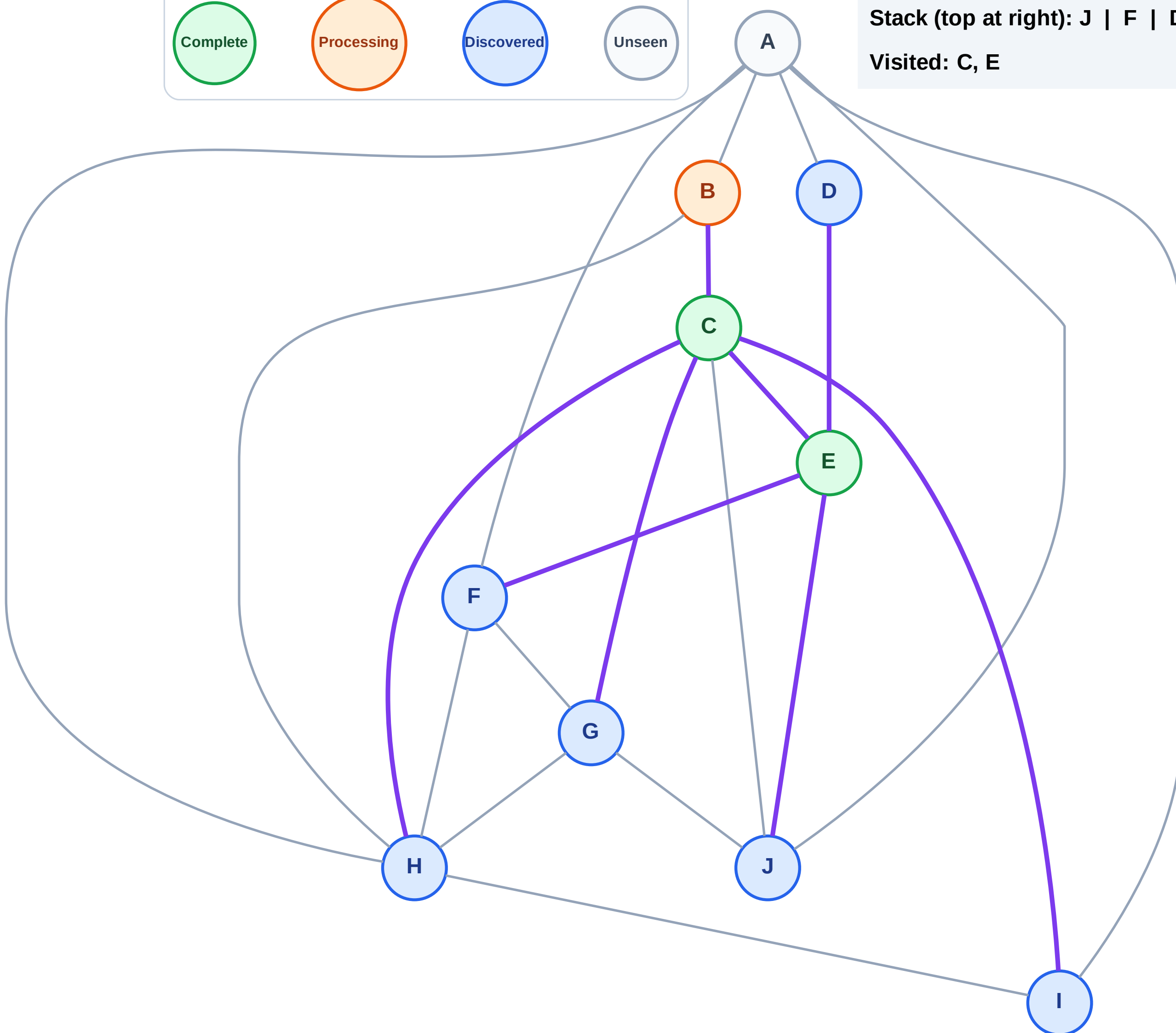
Discovered

Discovered

Unseen

Unseen

Visited: C, E



DFS Traversal

Step 7: Discover A from B

Legend

Complete

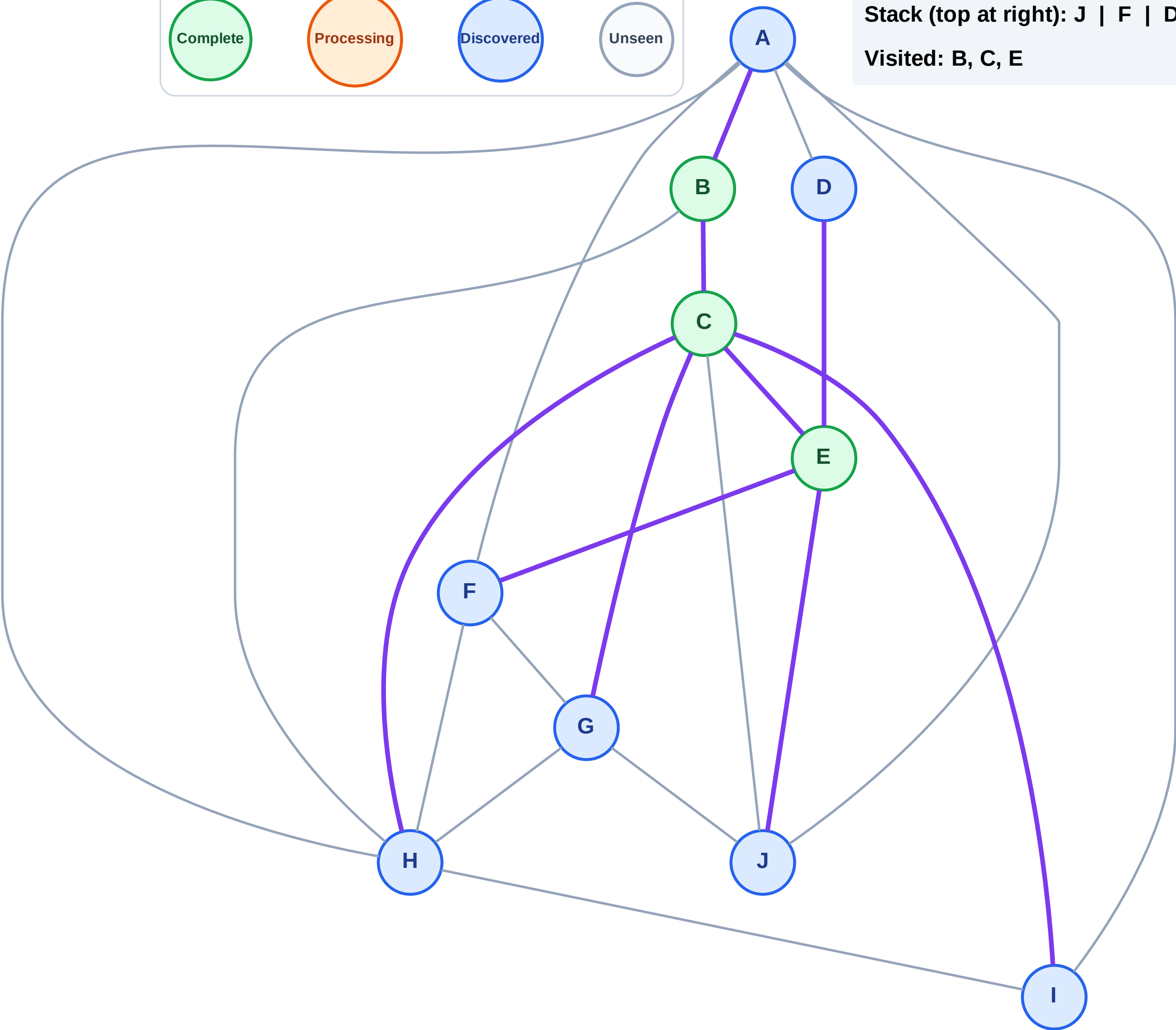
Processing

Discovered

Unseen

Stack (top at right): J | F | D | I | H | G | A

Visited: B, C, E



DFS Traversal

Step 8: Process node A

Legend

Complete

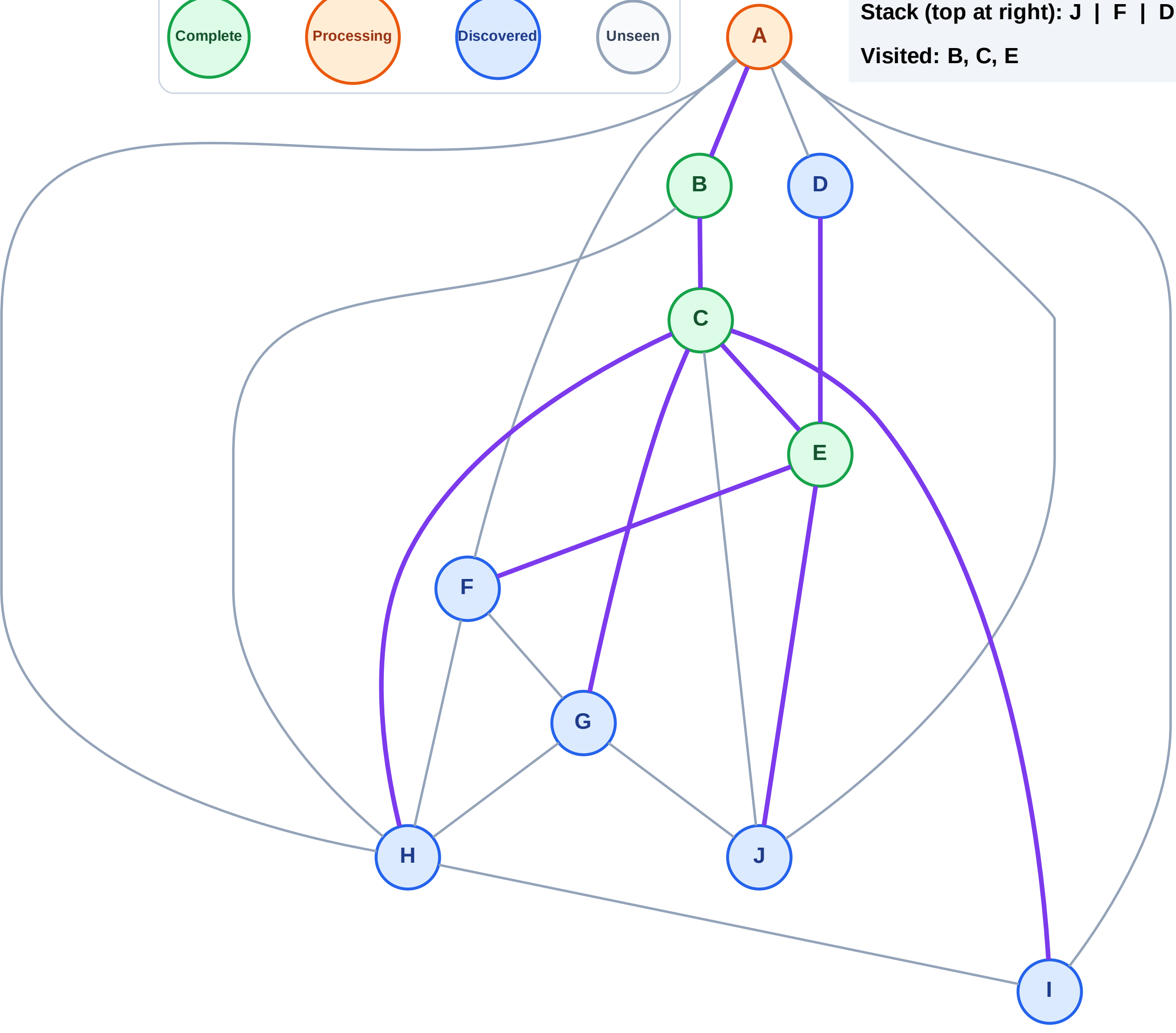
Processing

Discovered

Unseen

Stack (top at right): J | F | D | I | H | G

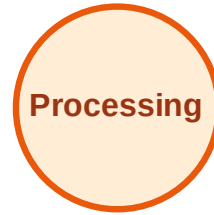
Visited: B, C, E



DFS Traversal

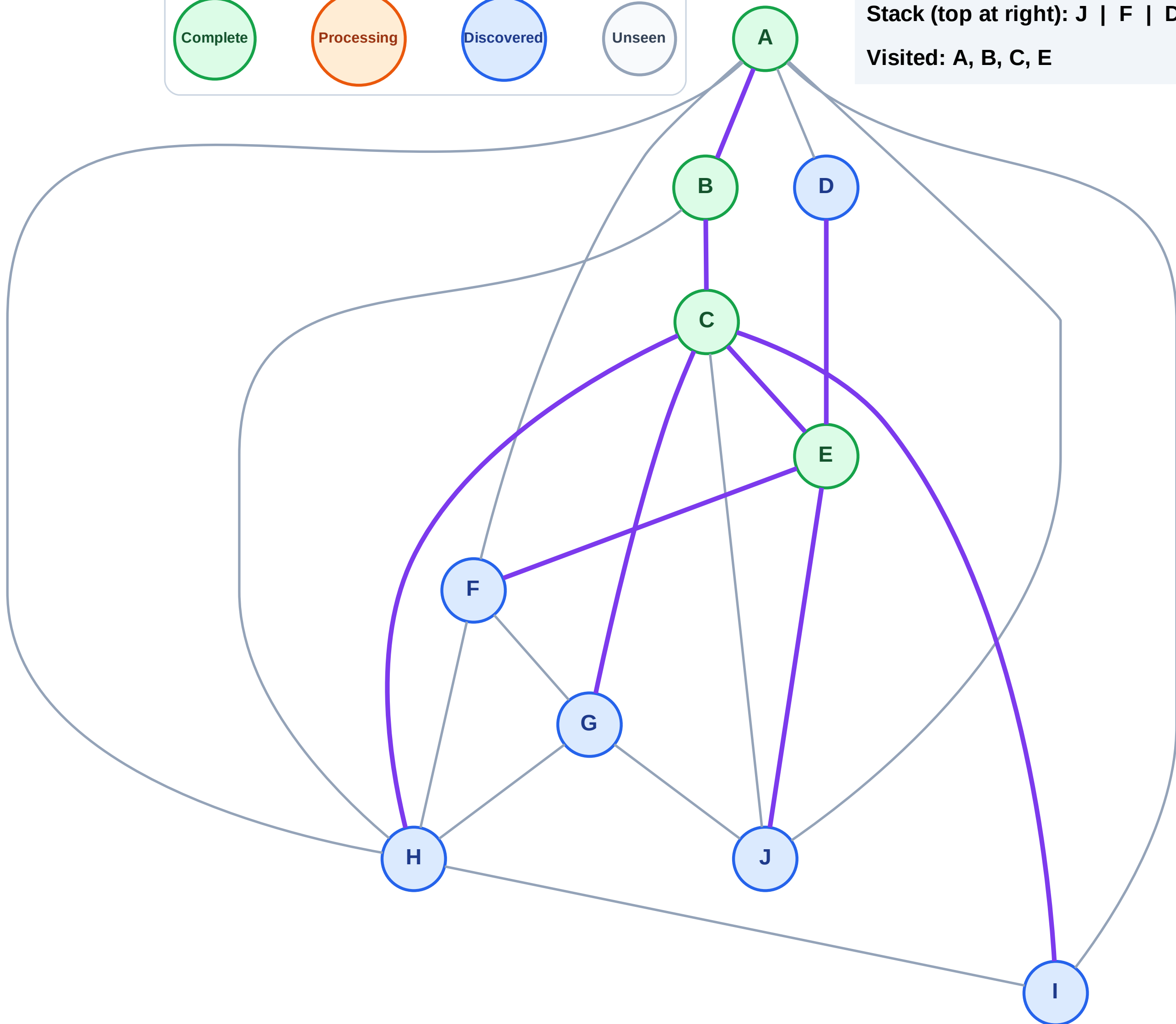
Step 9: No new neighbors from A

Legend



Stack (top at right): J | F | D | I | H | G

Visited: A, B, C, E



DFS Traversal

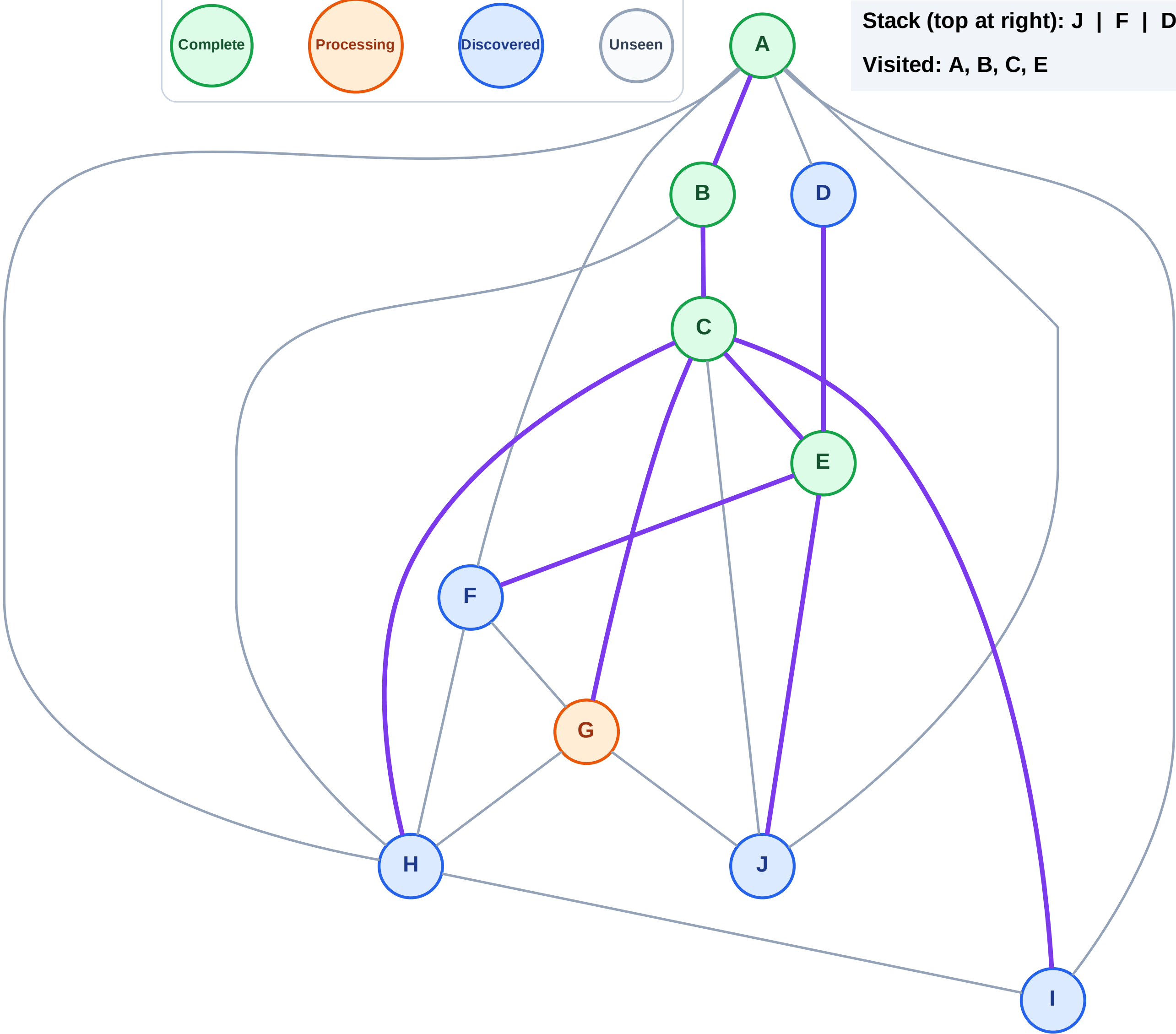
Step 10: Process node G

Legend



Stack (top at right): J | F | D | I | H

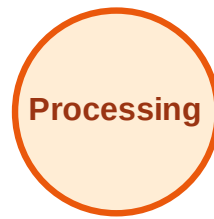
Visited: A, B, C, E



DFS Traversal

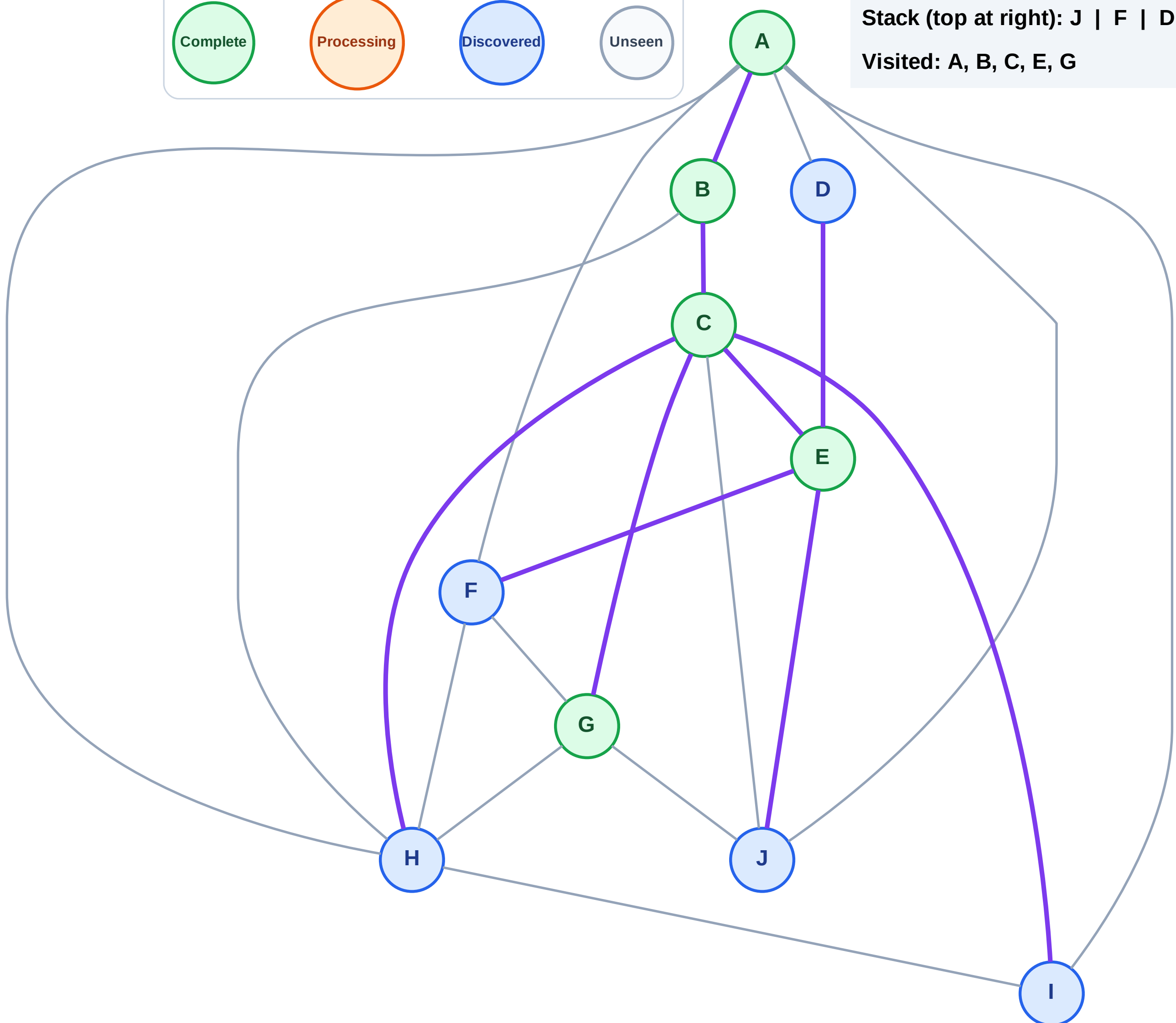
Step 11: No new neighbors from G

Legend



Stack (top at right): J | F | D | I | H

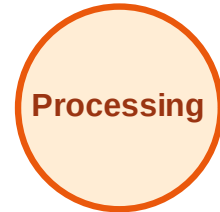
Visited: A, B, C, E, G



DFS Traversal

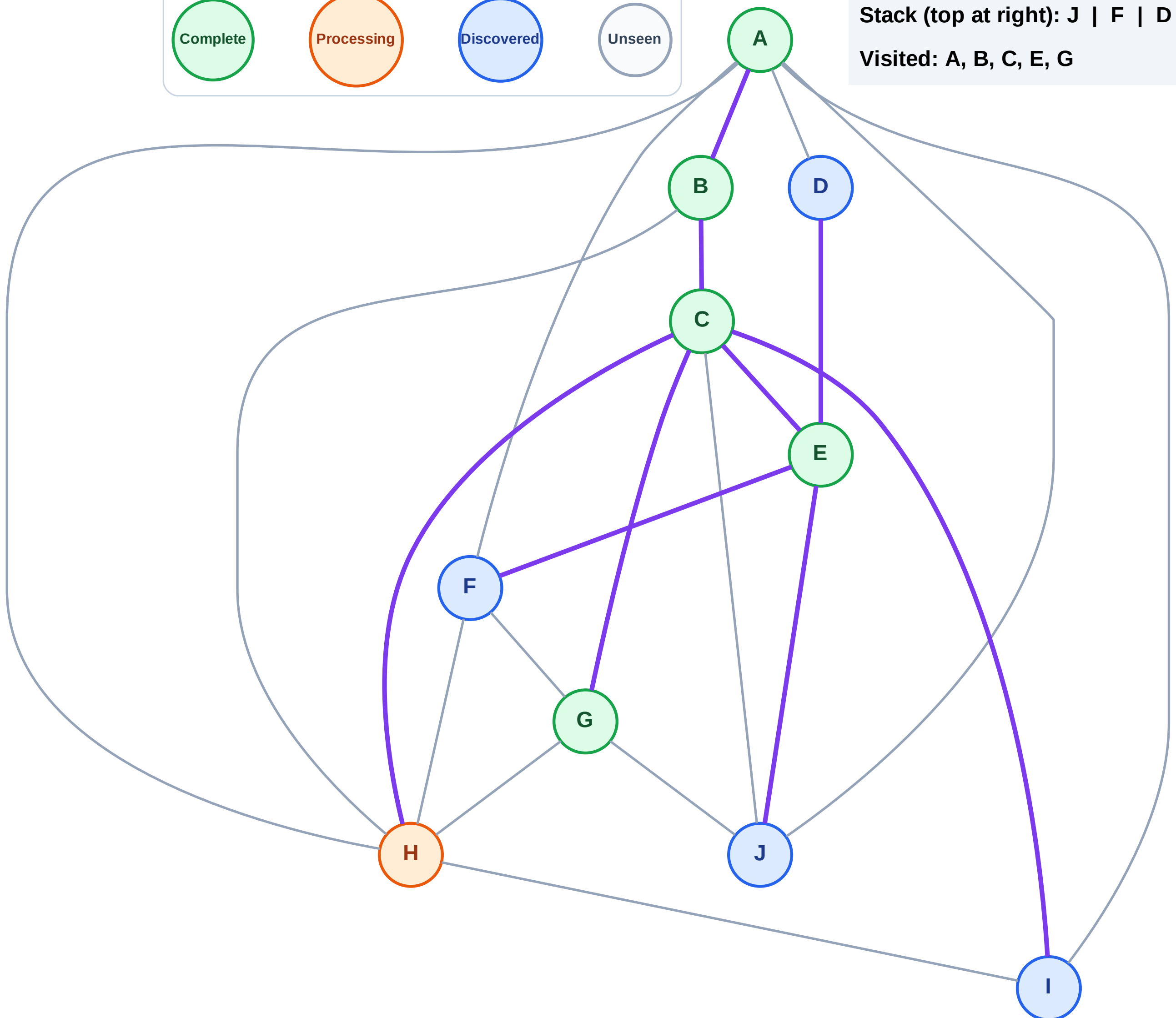
Step 12: Process node H

Legend



Stack (top at right): J | F | D | I

Visited: A, B, C, E, G



DFS Traversal

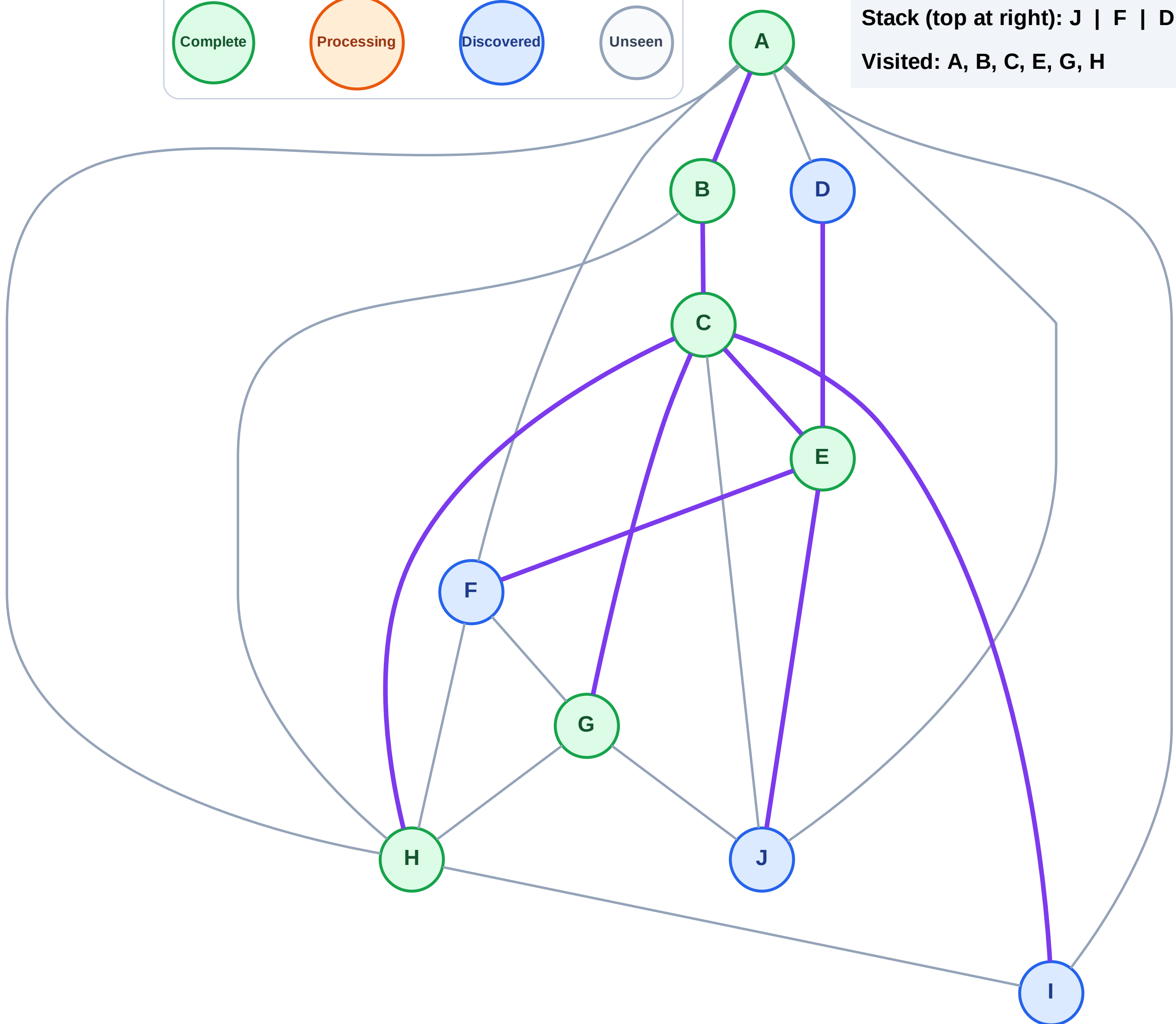
Step 13: No new neighbors from H

Legend



Stack (top at right): J | F | D | I

Visited: A, B, C, E, G, H



DFS Traversal

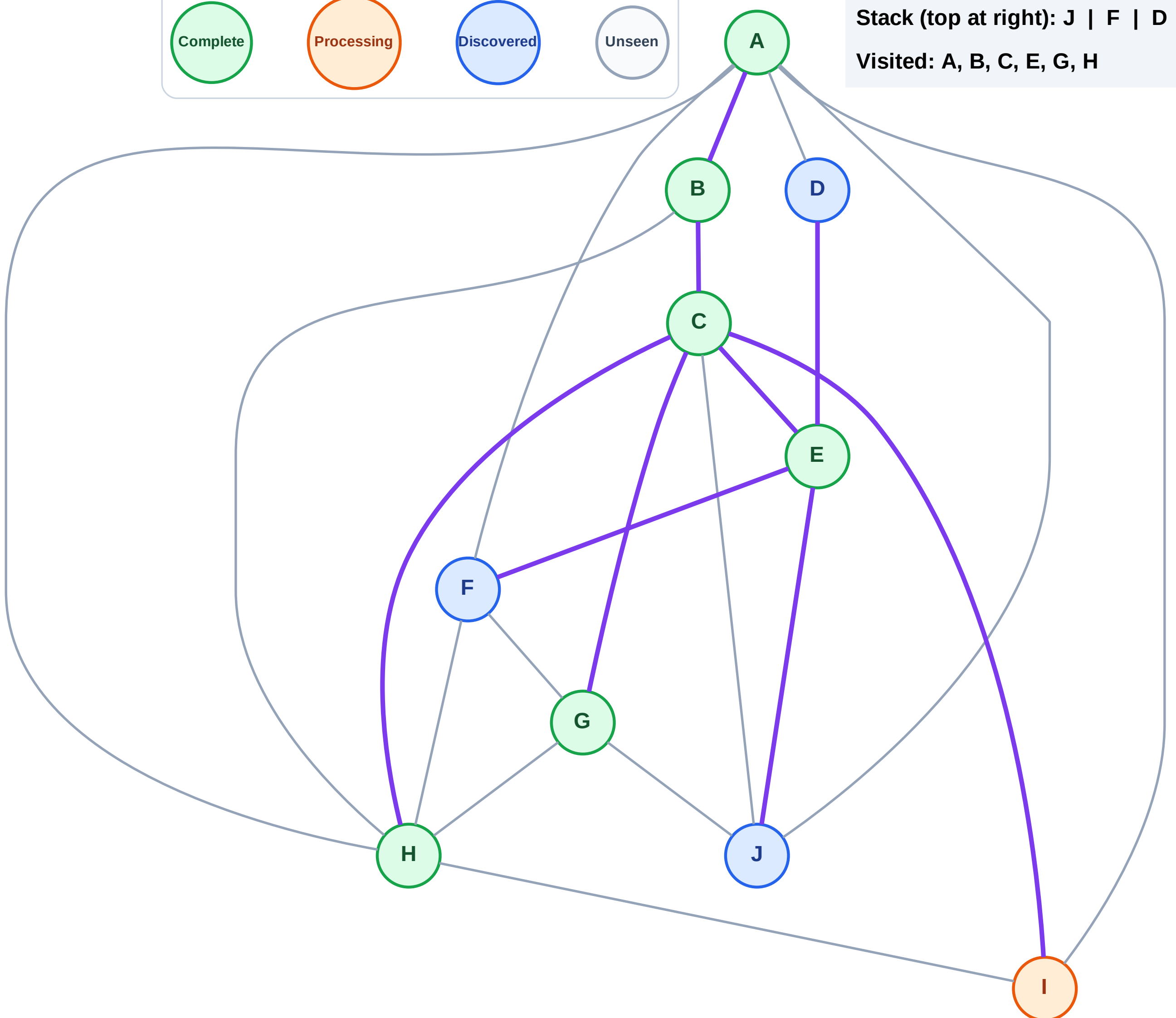
Step 14: Process node I

Legend



Stack (top at right): J | F | D

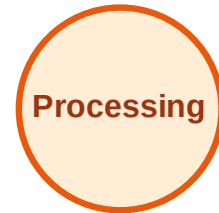
Visited: A, B, C, E, G, H



DFS Traversal

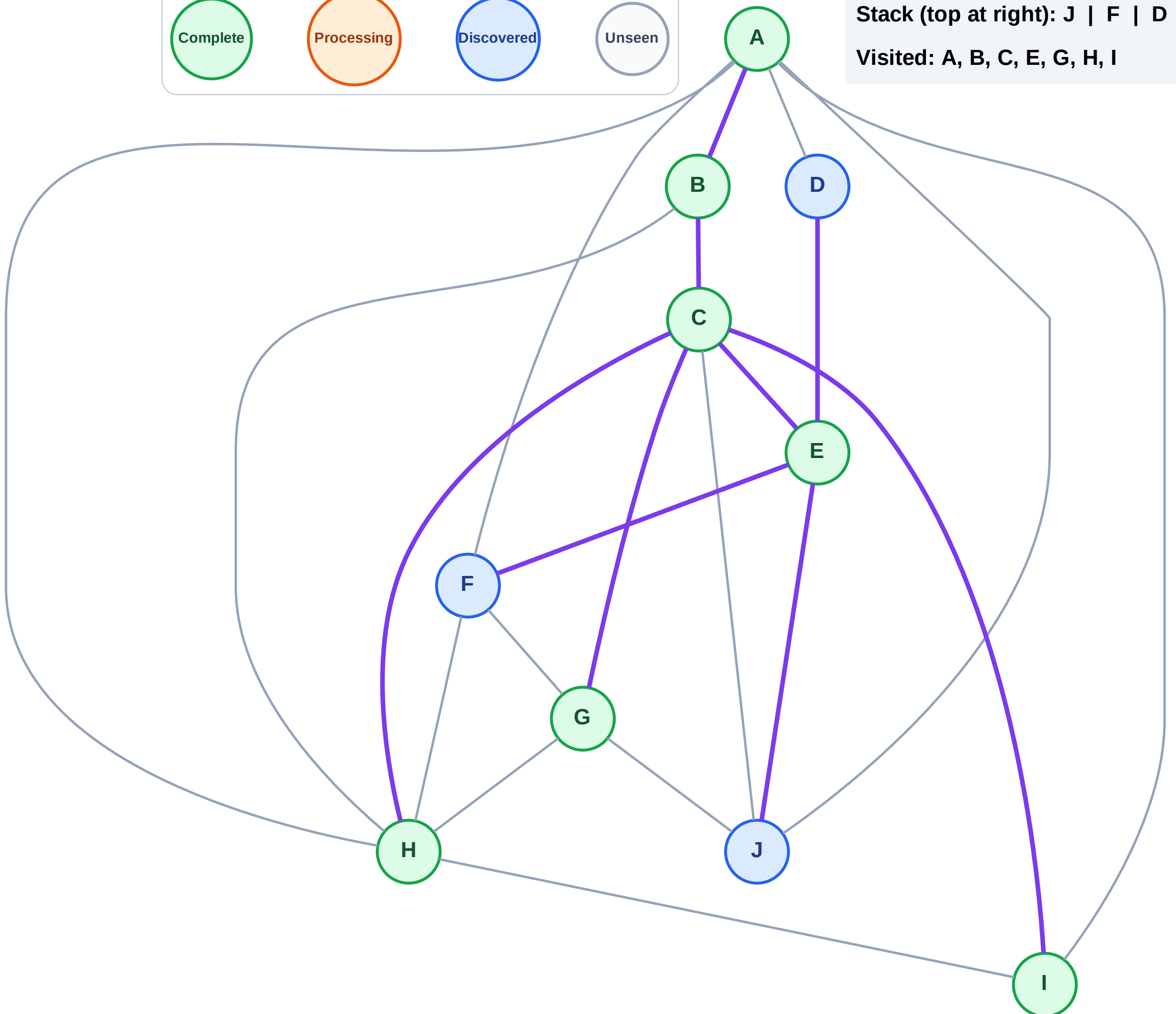
Step 15: No new neighbors from I

Legend



Stack (top at right): J | F | D

Visited: A, B, C, E, G, H, I



DFS Traversal

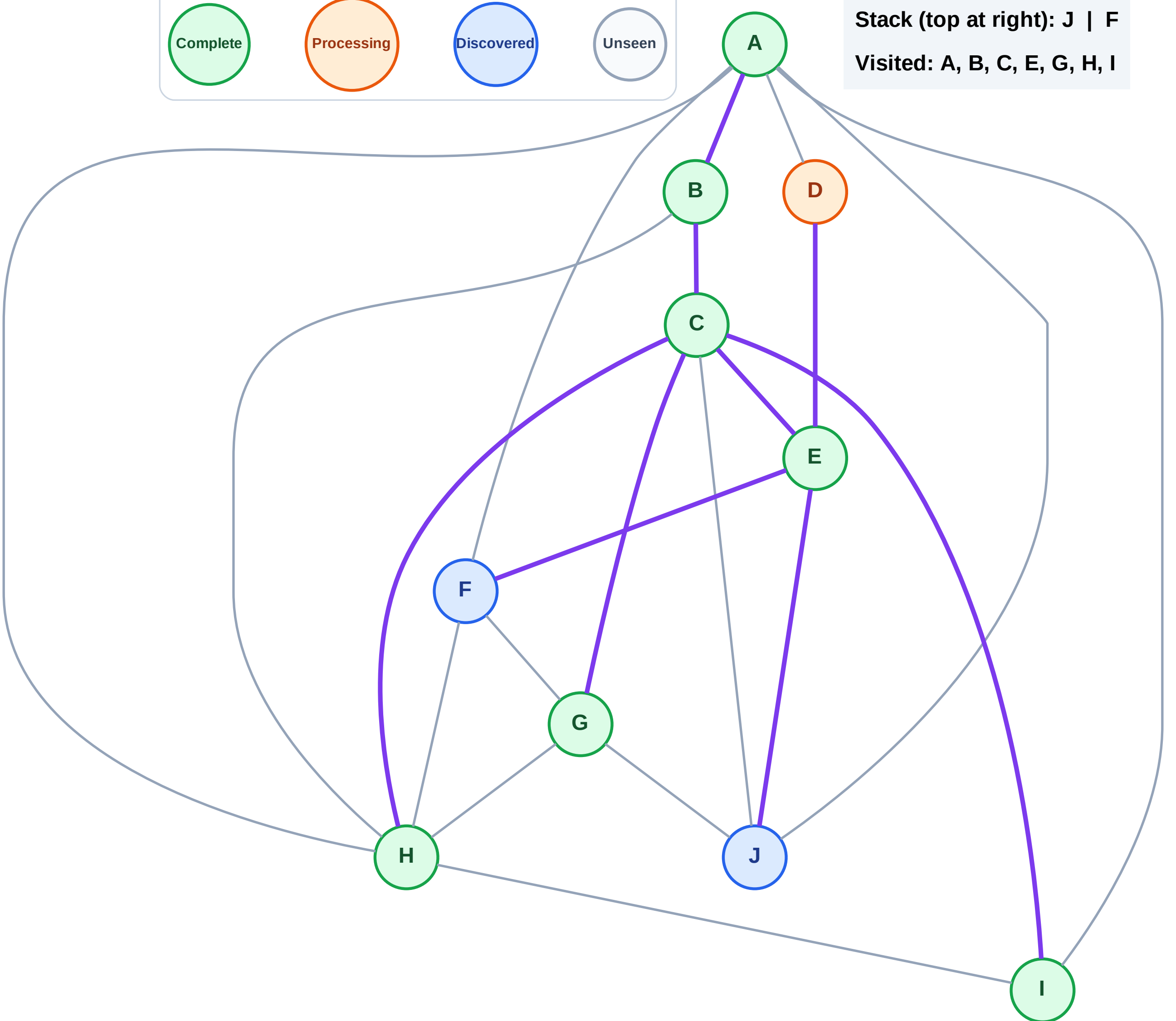
Step 16: Process node D

Legend



Stack (top at right): J | F

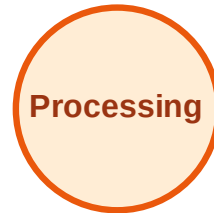
Visited: A, B, C, E, G, H, I



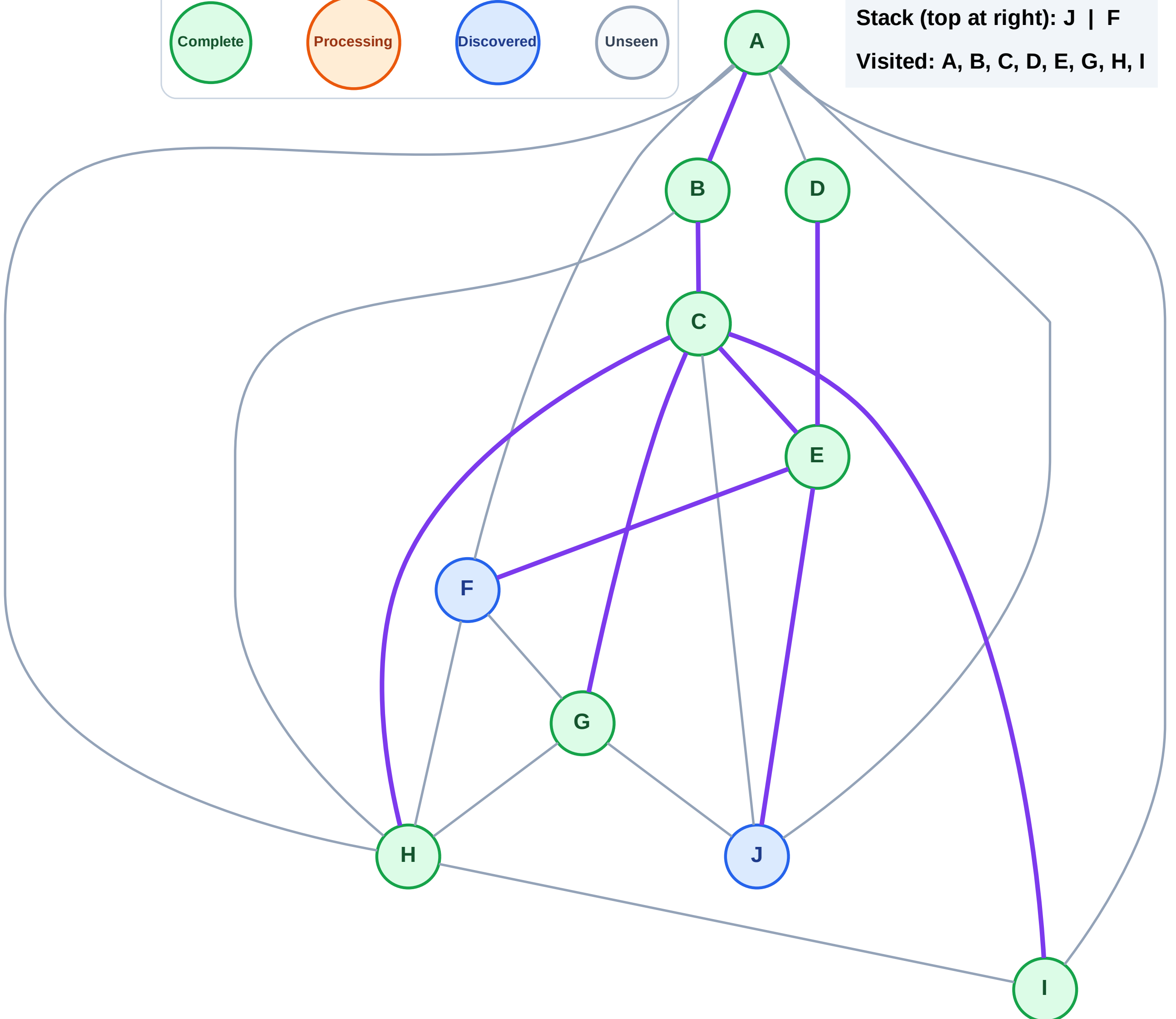
DFS Traversal

Step 17: No new neighbors from D

Legend



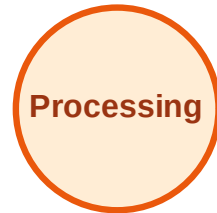
Stack (top at right): J | F
Visited: A, B, C, D, E, G, H, I



DFS Traversal

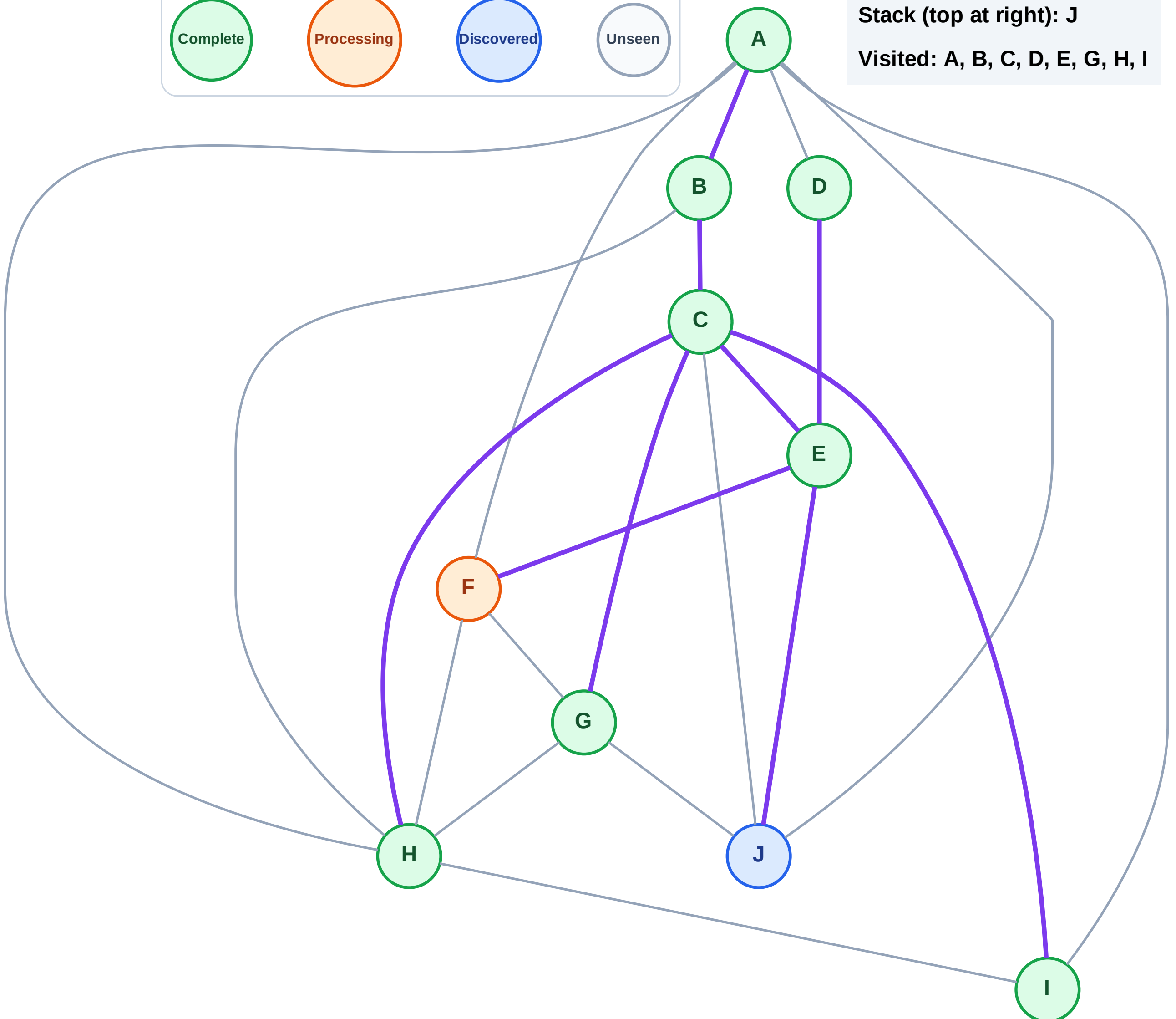
Step 18: Process node F

Legend



Stack (top at right): J

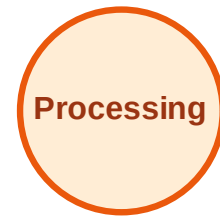
Visited: A, B, C, D, E, G, H, I



DFS Traversal

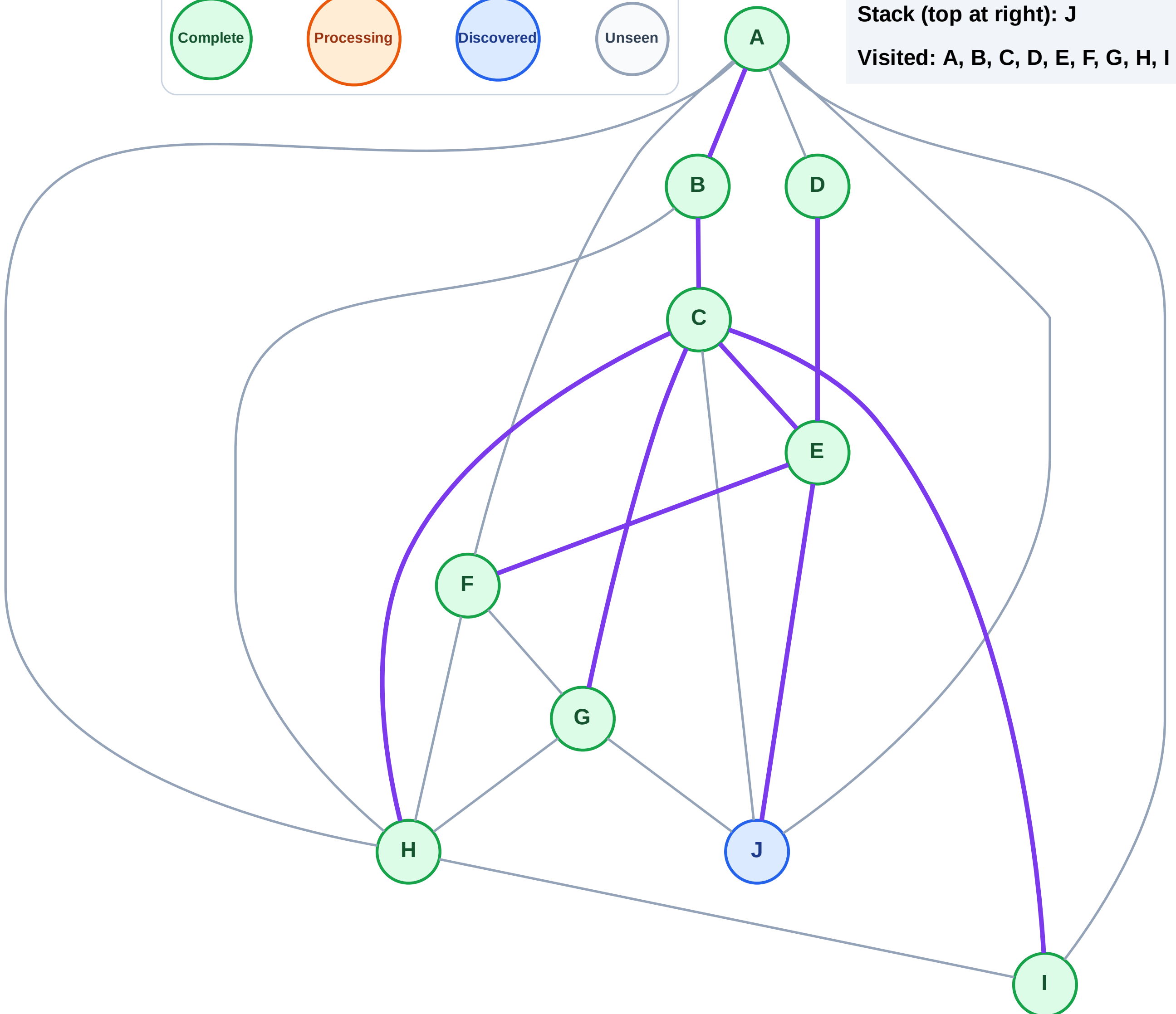
Step 19: No new neighbors from F

Legend



Stack (top at right): J

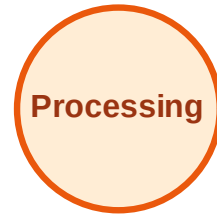
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

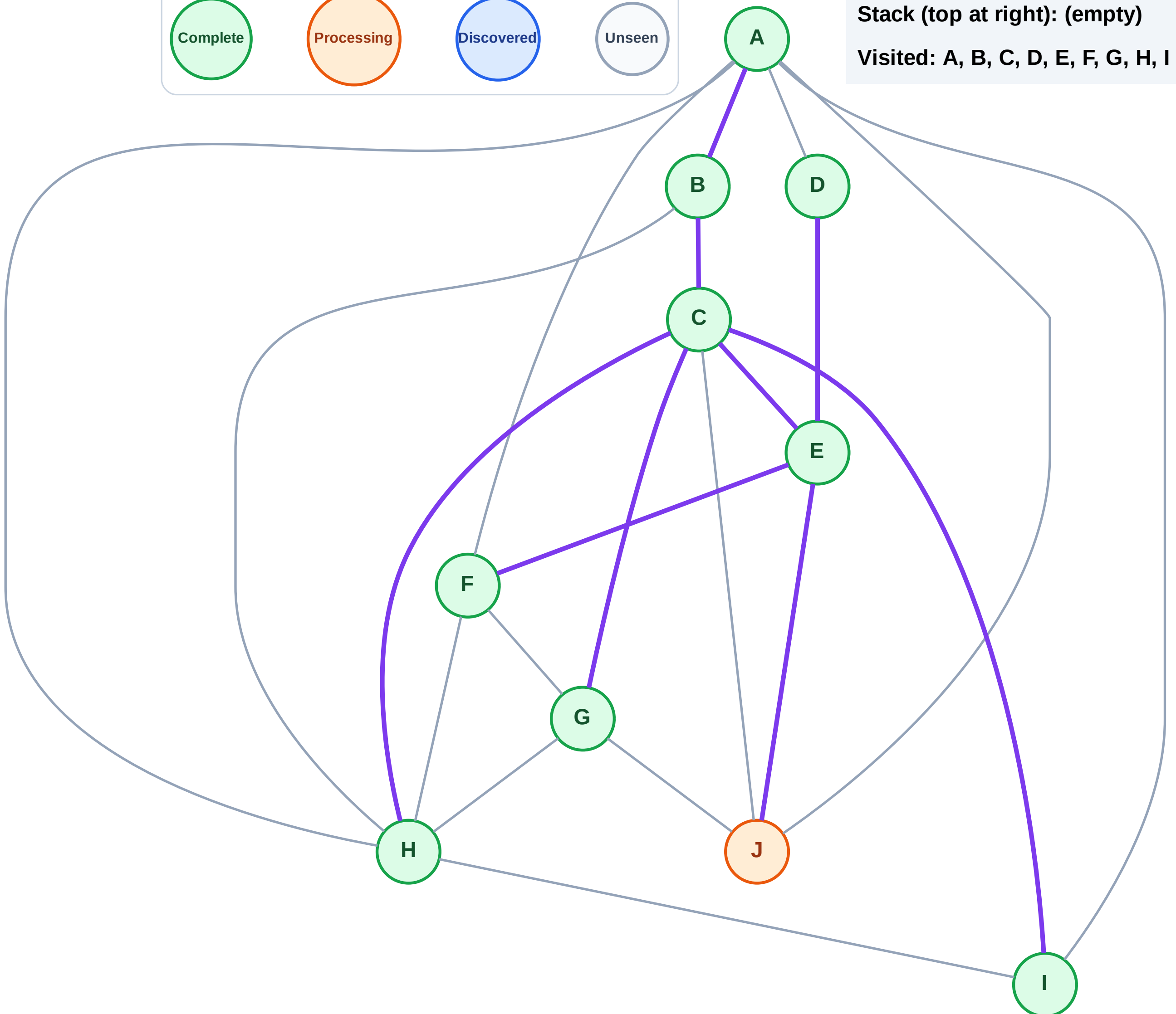
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

